Characterizing ocean waves and sea states

MEC_5EO04_TA / MEC_53456_EP – Class 1
September 25, 2026

Marissa L. Yates

marissa.yates@enpc.fr

Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09
2. Linear wave theory	Characterizing offshore waves	M. Yates	09/10
3. Nearshore wave propagation	Calculating extreme wave conditions	M. Yates	16/10
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	23/10
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	06/11
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	04/12
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12
10. Final project presentations 13h30 - 17h15			***18/12***

Objectives

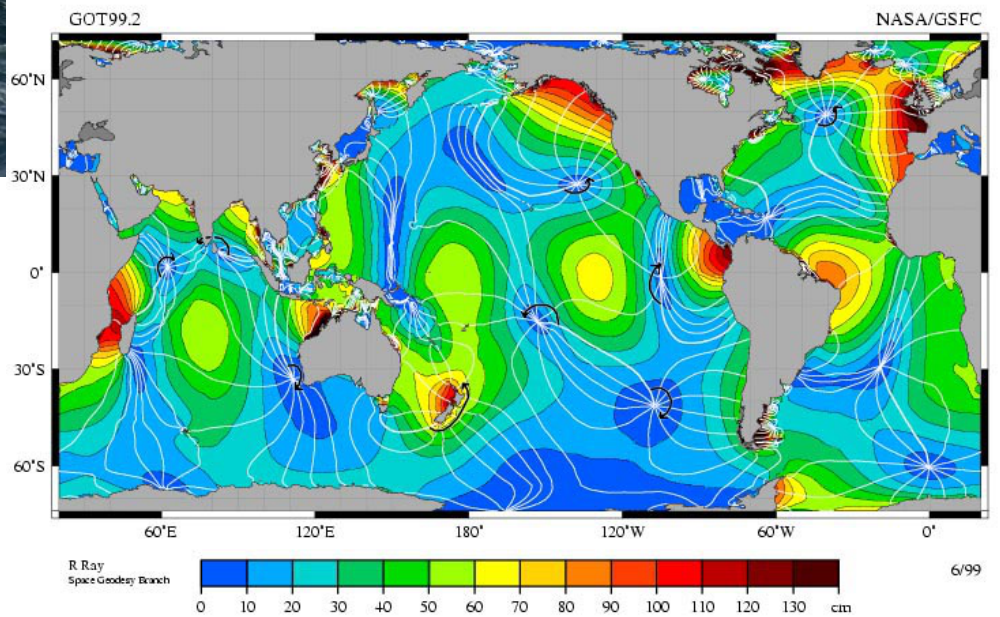
- Introduce wave and sea state terminology
- Describe how to characterize waves
- Describe how to characterize sea states:
  - Statistical approach
  - Spectral approach
- Introduce wave observation techniques
- Present existing (observational and model) wave databases

Free surface ocean waves

Capillary waves

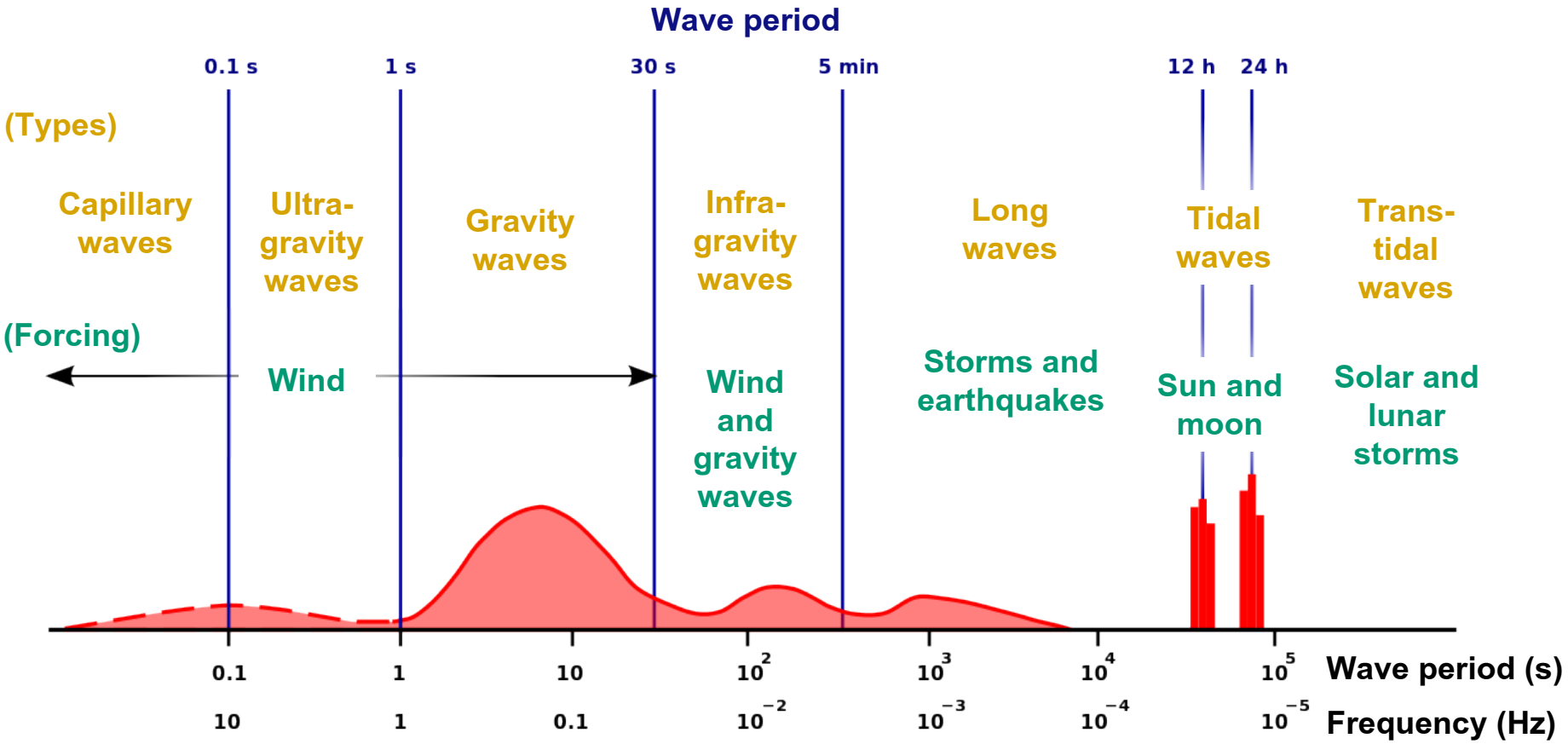
Gravity waves

Tidal waves



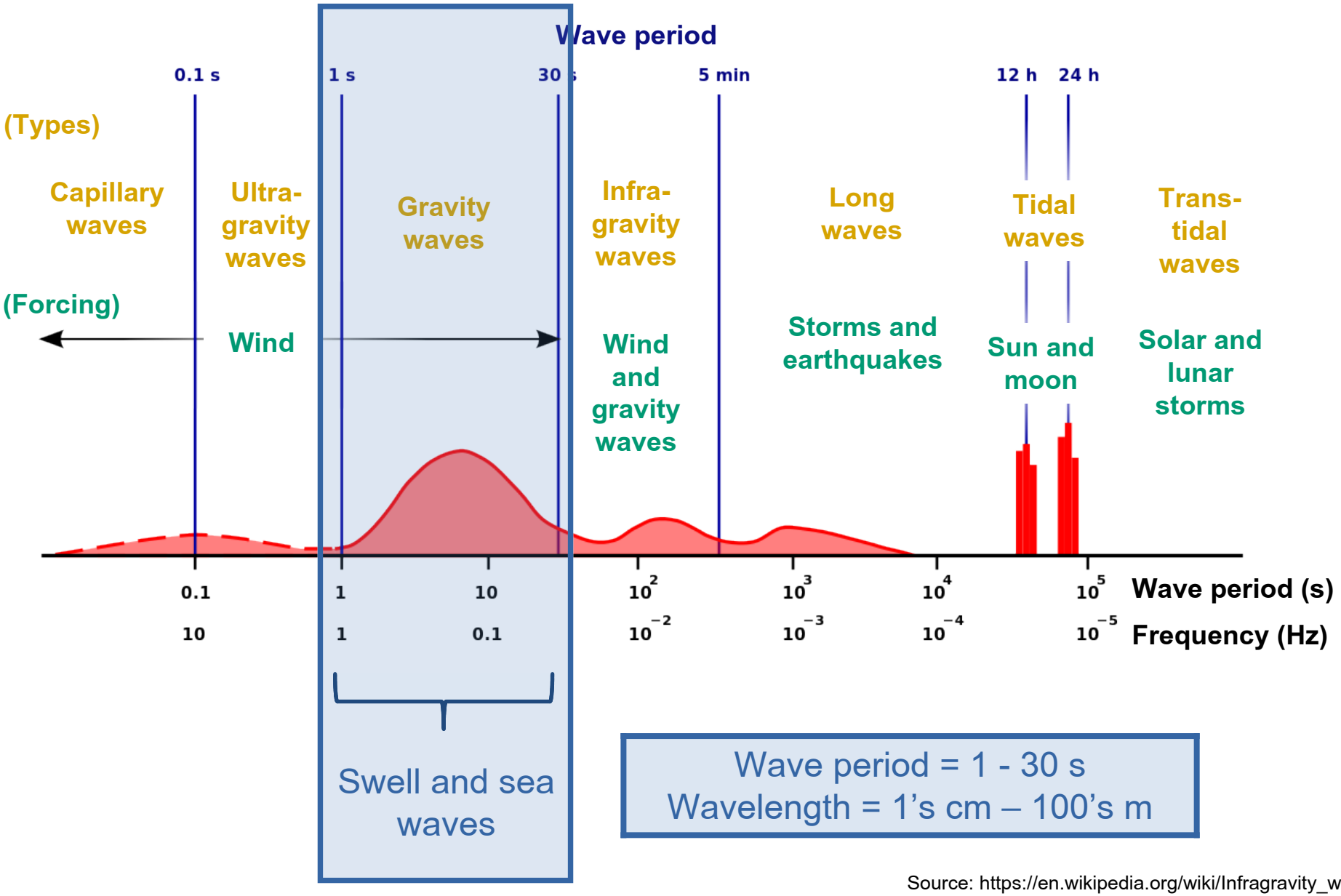
- Different spatial and temporal scales
- Different physical generation mechanisms
- Superposition and interactions

Free surface ocean waves



Source: https://en.wikipedia.org/wiki/Infragravity_wave

Free surface ocean waves



Source: https://en.wikipedia.org/wiki/Infragravity_wave

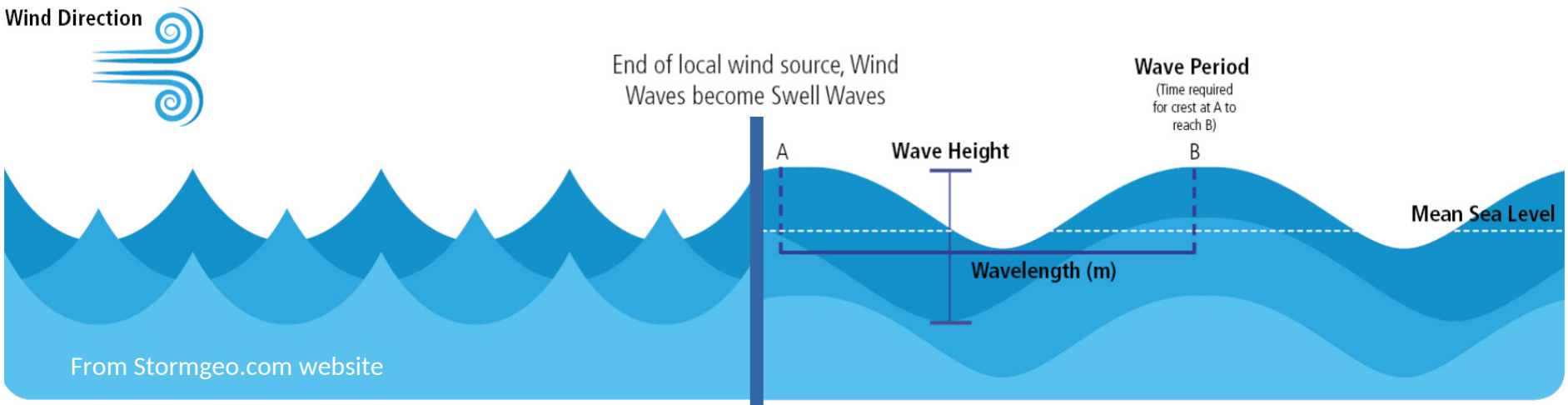
Definitions

WIND WAVES

Wind Waves are generated by immediate local wind. They are not self-sustaining and will die out when the wind stops.

SWELL WAVES

Swell Waves are self-sustaining and generated by energy beneath the ocean's surface, no longer needing local wind.



- Waves observed in the zone of wave generation (also called sea waves)
- Irregular in wave height and period, direction close to wind direction (with some angular spreading)
- Characterized by short periods = 2-8 s

- Waves observed far from the generation zone / or after the wind has ceased
- More regular in wave height, period, and direction
- Characterized by longer periods = 10-30 s

Definitions

WIND WAVES

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- More regular in wave height, period, and direction
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Definitions

WIND WAVES

+

SWELL WAVES

=

Sea state

- Local wave conditions composed of the superposition of waves of different origin (swell and sea/wind waves)
- Characteristic length of time used to describe sea states : 30 min – 3 h

How are waves characterized?

Time

100 yrs

10 yrs

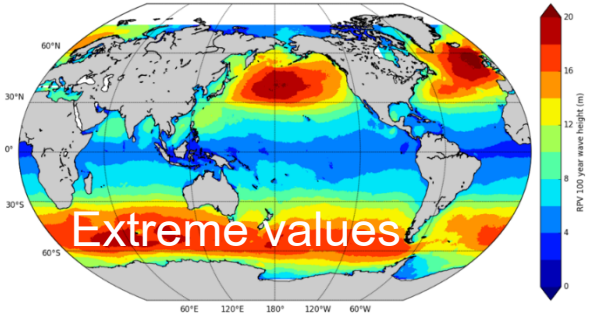
yrs

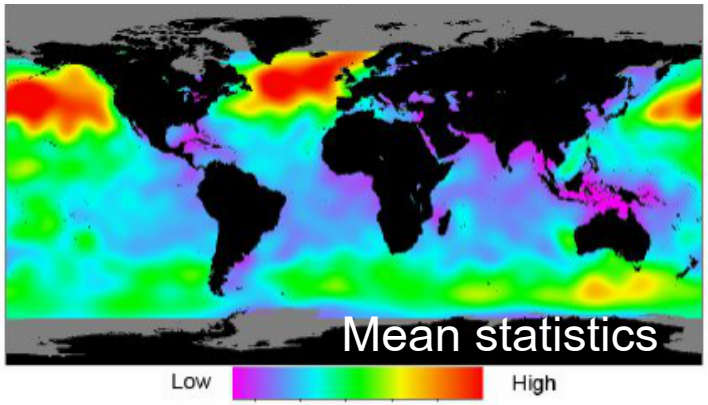
months

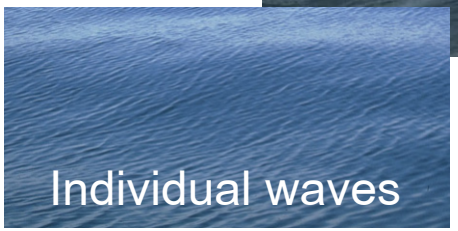
hrs

mins

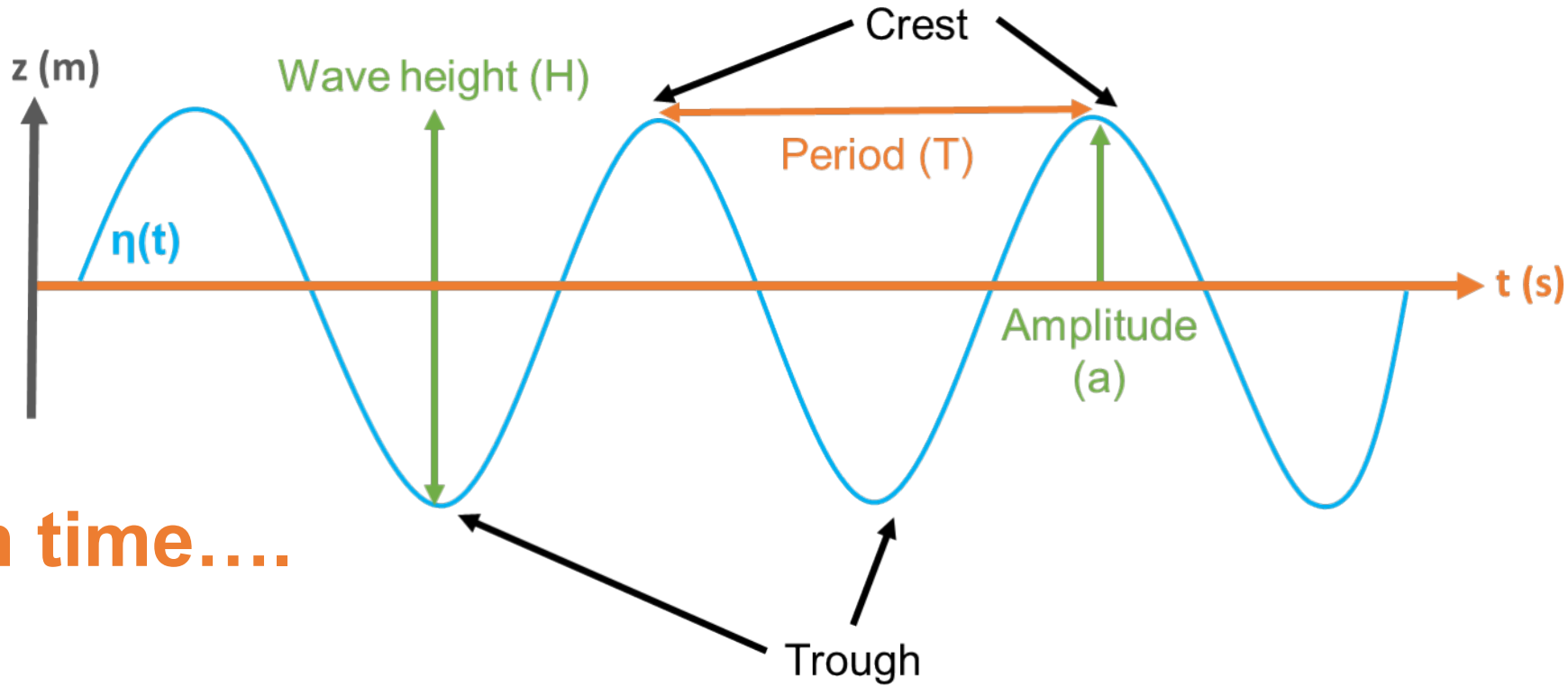
secs







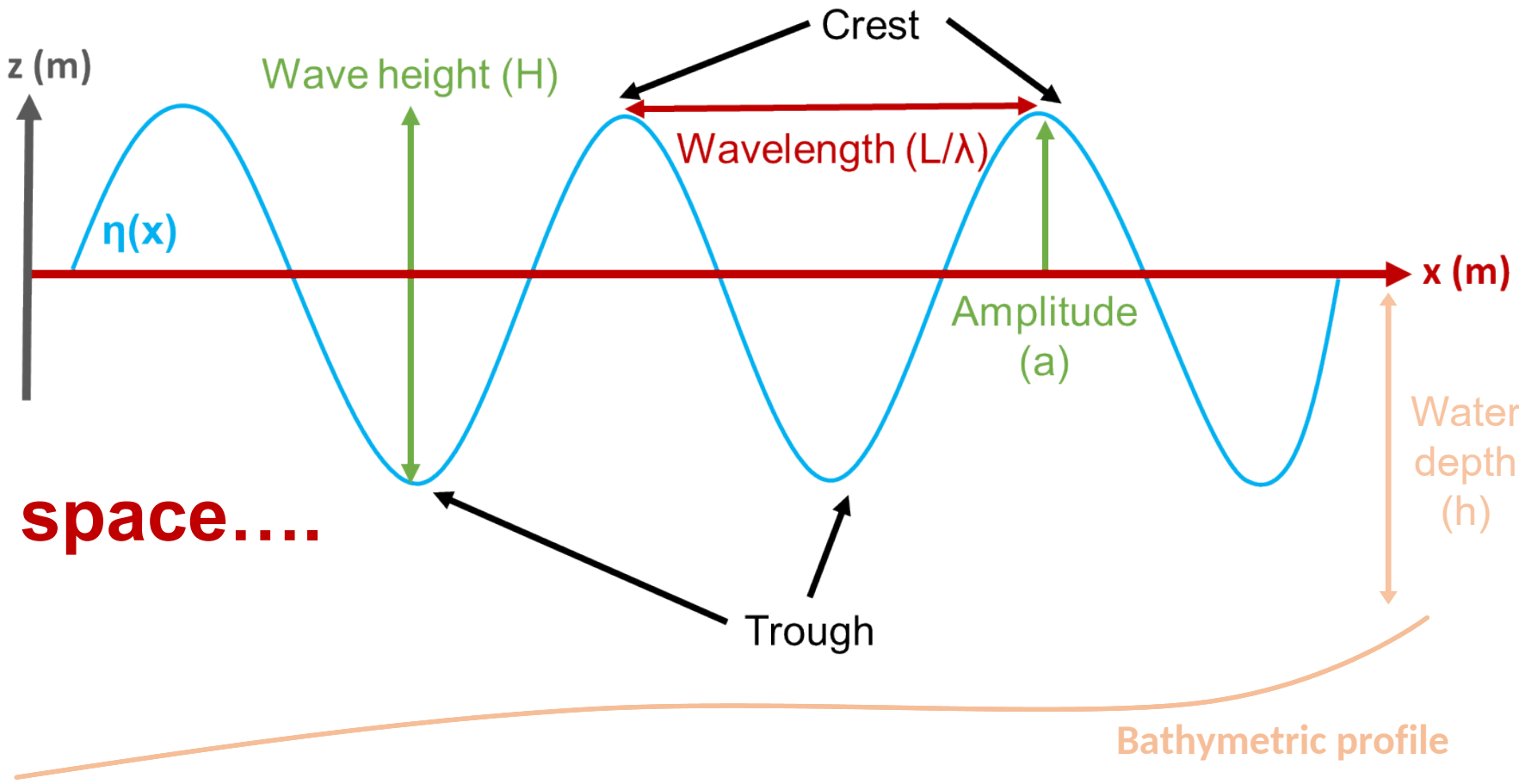
How are waves characterized?



In time….

Individual waves

How are waves characterized?

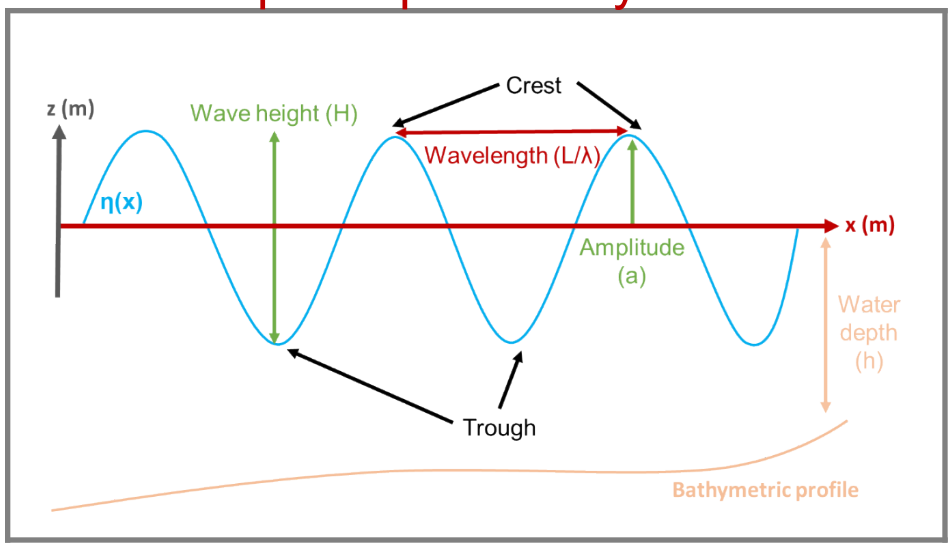


In space….

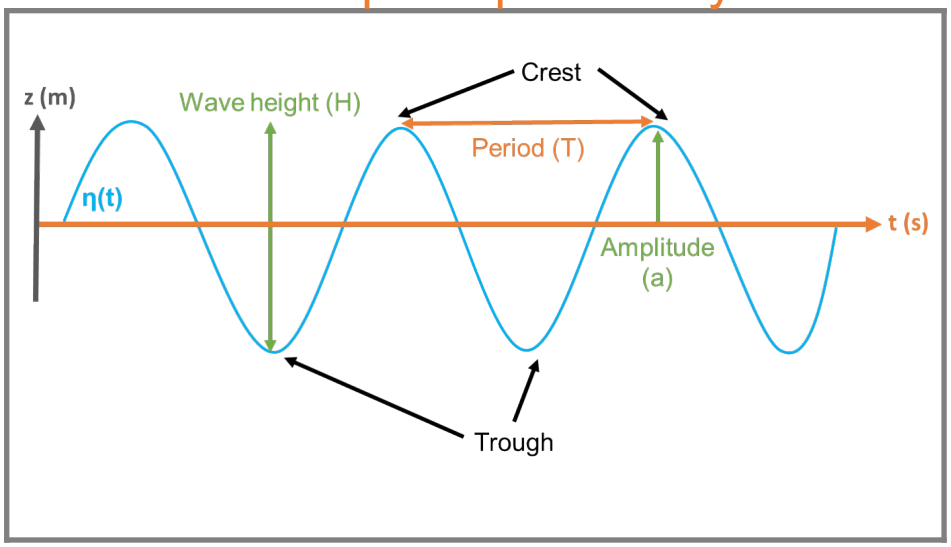
Individual waves

Wave parameters

Spatial periodicity



Temporal periodicity



Wavelength = L or λ (m)
Wave number = k = 2π/L (rad/m)

Wave period= T (s)
Frequency = f = 1/T (Hz)
Frequency = ω = 2π/T (rad/s)

Link between the spatial and temporal domains:

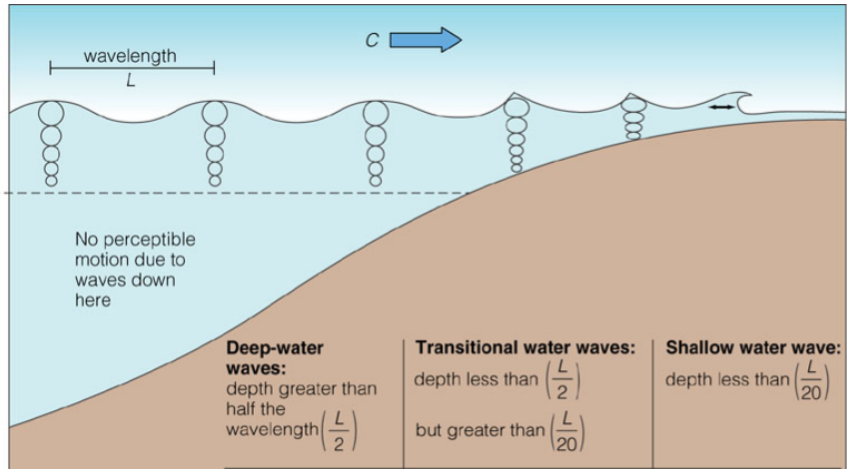
Dispersion relation

$$\omega^2 = gk\tanh(kh)$$

Nondimensional numbers used to characterize waves

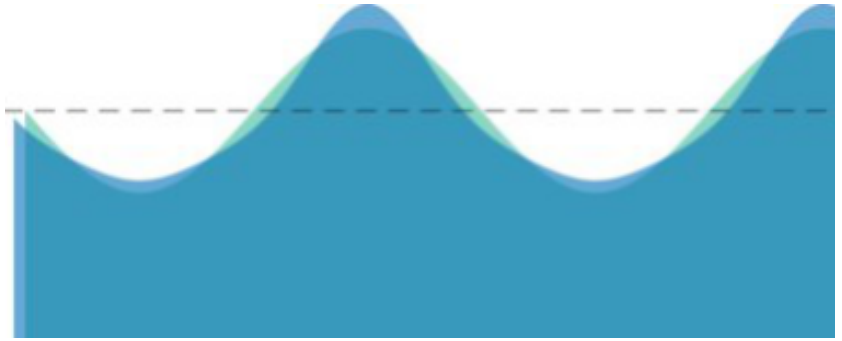
- **Dispersion: effects of *h***

  Do waves « feel » the bottom?



- **Nonlinearity: effects of *H***

  Are waves a « large » perturbation of the free surface?



Wave propagation speed: $C = \frac{L}{T} = f(h, T, H)$

Nondimensional numbers used to characterize waves

- **Dispersion: effects of *h***

  Do waves « feel » the bottom?

  Relative water depth:

  h/L or $\mu = kh$

  Nondimensional period:

  $T(g/h)^{1/2}$

- **Nonlinearity: effects of *H***

  Are waves a « large » perturbation of the free surface?

  Steepness (deepwater):

  $s = H/L$ or $\epsilon = ka$

  Relative wave height (shallow water):

  H/h

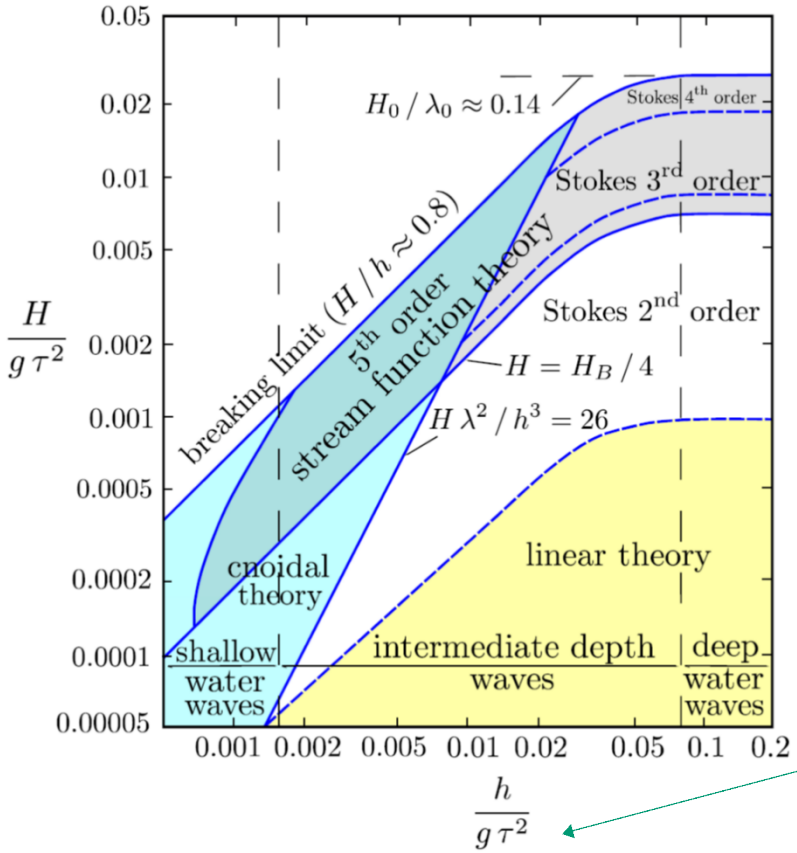
Impacts on wave shape and propagation speed

Nondimensional numbers used to characterize waves

- **Importance of nonlinearity vs dispersio**
  Ursell number = nonlinearity/(dispersion)^3

$$Ur = \frac{HL^2}{h^3}$$

Nonlinearity parameter



Dispersion parameter

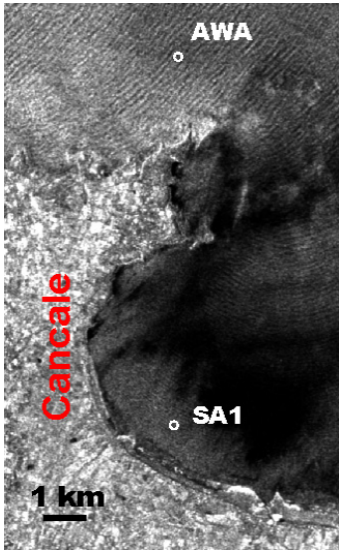
Wave observations

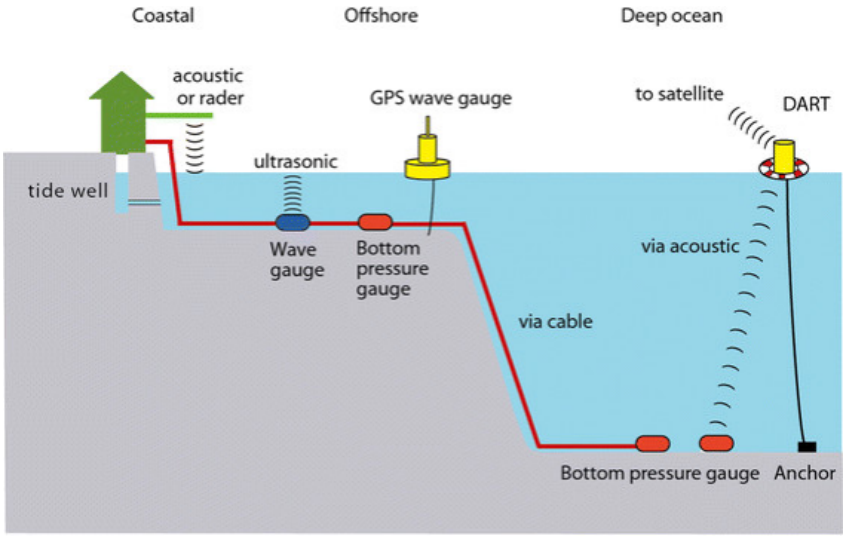
How can we characterize waves?
How do we measure waves?

https://berowraba

Measuring waves

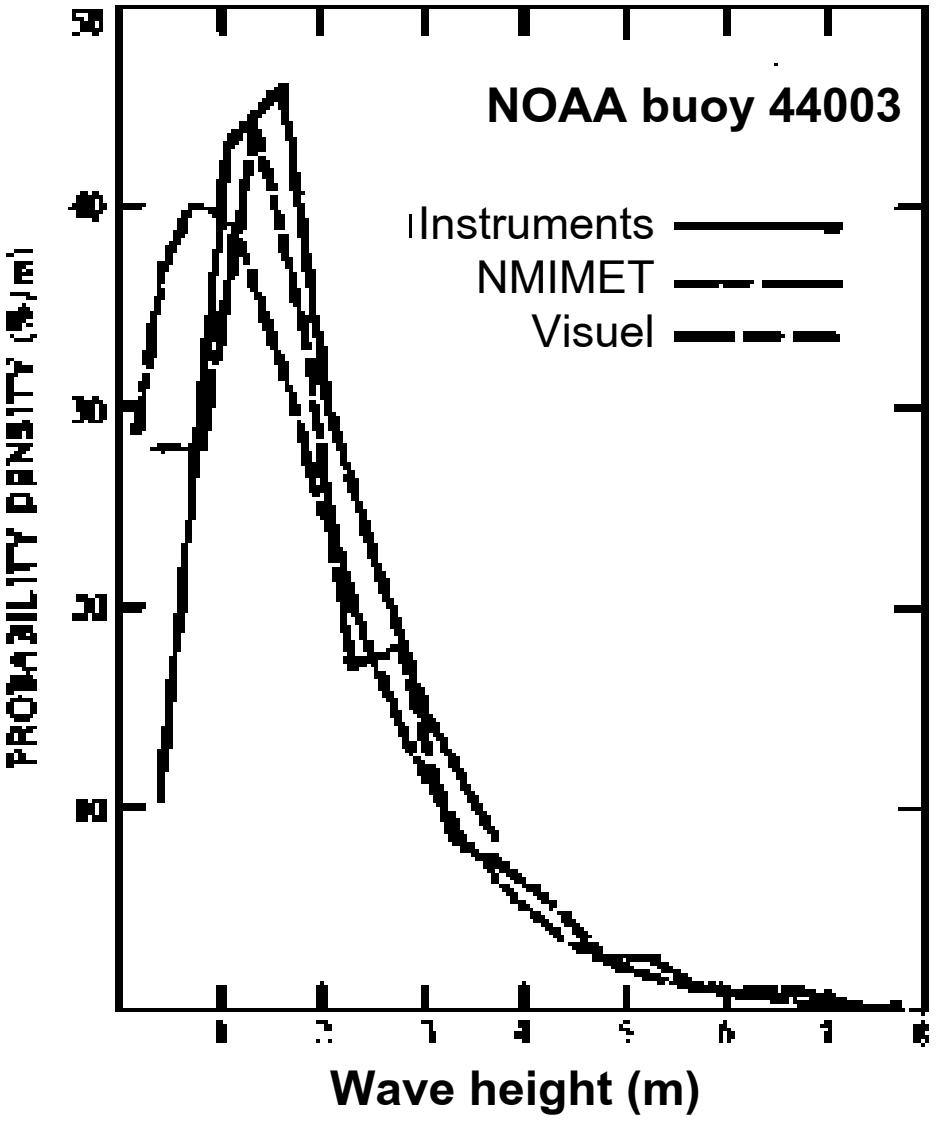
- **Ship and platform observations**
  - Visual observations (historically)
- **In situ observations**
  - Wavestaff
  - Buoys
  - Pressure sensors
- **Remote sensing observations**
  - Satellite altimetry
  - Radar imagery
  - Lidar
  - Stereovideo
  - Drones





Measuring waves

- **Ship and platform observations**
  - Visual observations (historically)



http://www.globalwavestatisticsonline.com/Help/height_data.htm

Measuring waves

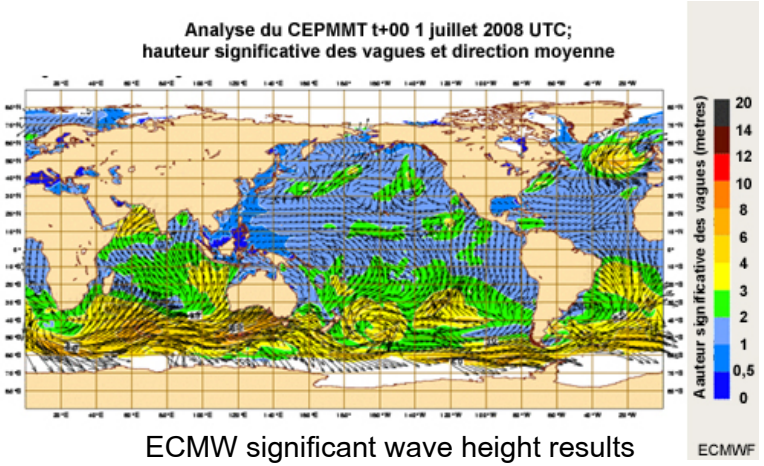
- **In situ observations**

  - Wave buoys: measure displacement/accelerations (3D) of the wave buoy (accelerometer or GPS)

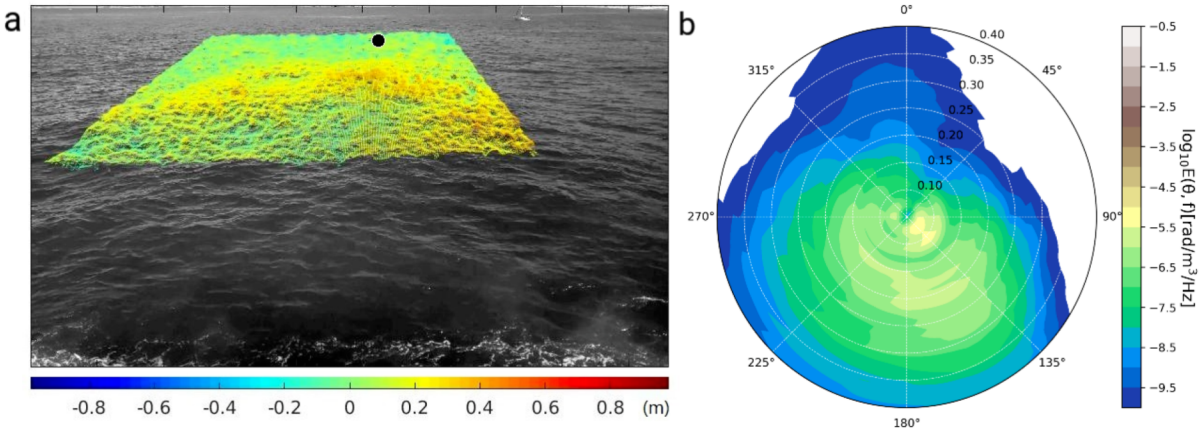
  - Pressure sensors: pressure change with passing wave (hydrostatic, or nonhydrostatic with nonlinear correction term)

Measuring waves

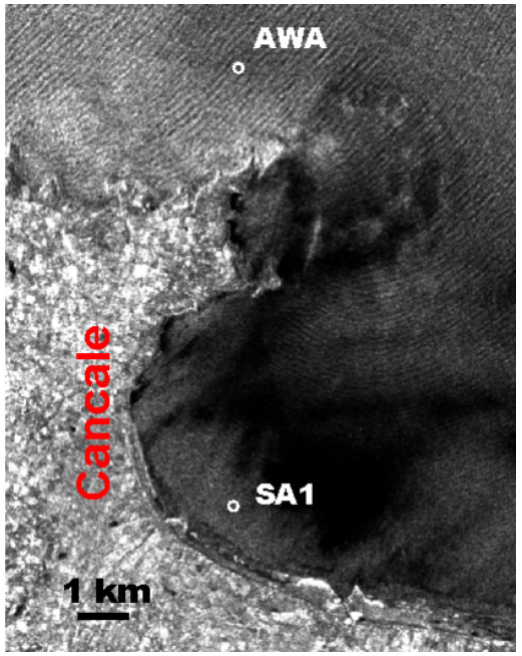
- **Remote sensing observations**
  - Satellite altimetry
  - Radar images (satellites or in situ)
  - Lidar (free surface position)
  - Stereovideo
  - Drones



ECMW significant wave height results
(https://www.eumetsat.int/jason_fr/print.htm)



Stereovideo measurements of free surface height
(Veira et al., 2020)



Synthetic aperature radar (SAR), from
ENVISAT satellite in March 2003 (SHOM)

Wave observations

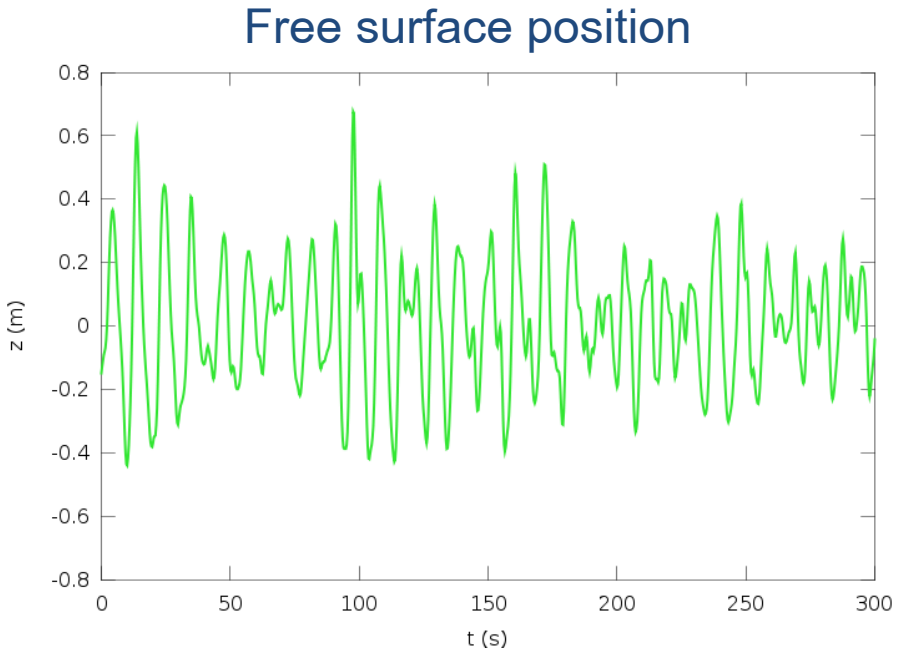
Linear waves (sinusoidal)
- regular waves
- defined by H,T

Sea states
- superposition of linear and nonlinear waves
- how can they be characterized?

https://berowrabaokyard.wordpress

Individual waves → **Sea states**

Wave observations



Deterministic approach:
Variations in time of the free surface position

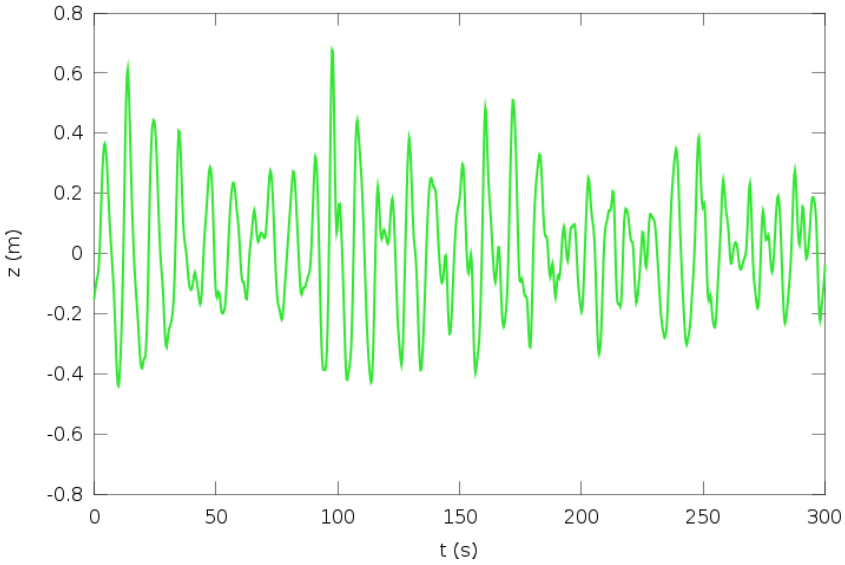
Phase-**resolving** wave models

Spectral or statistical approach:
Variations in time average wave characteristics

Phase-**averaged** wave models

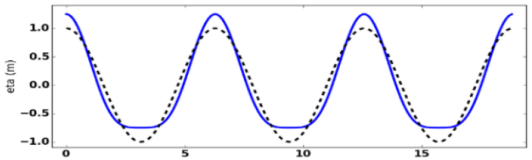
Deterministic approach

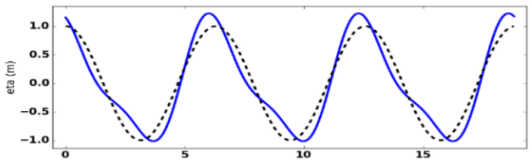
Temporal variations of the free surface position : calculate moments



Averaging operator: $< f >= \frac{1}{N} \int_0^N f(t)dt$

- Mean $< \eta >= \frac{1}{N} \int_0^N \eta(t)dt$
- Variance $m_0 = \sigma^2 = \left\langle (\eta - \langle \eta \rangle)^2 \right\rangle$
- Skewness (horizontal asymmetry) $\lambda_3 = \frac{\left\langle (\eta - \langle \eta \rangle)^3 \right\rangle}{m_0^{3/2}}$
- Kurtosis (vertical asymmetry) $\lambda_4 = \frac{\left\langle (\eta - \langle \eta \rangle)^4 \right\rangle}{m_0^2}$





For linear (Gaussian) sea states:

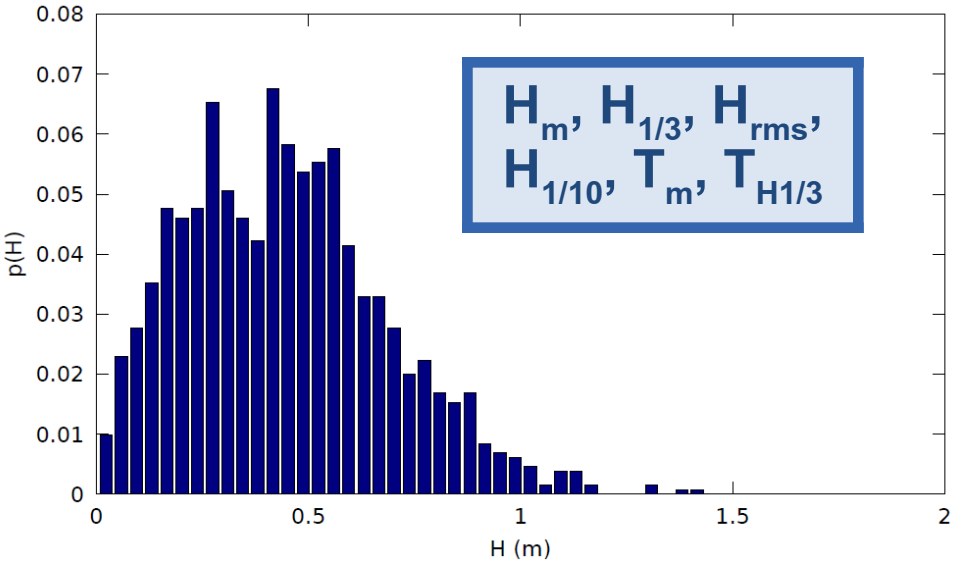
$\lambda_3 = 0$

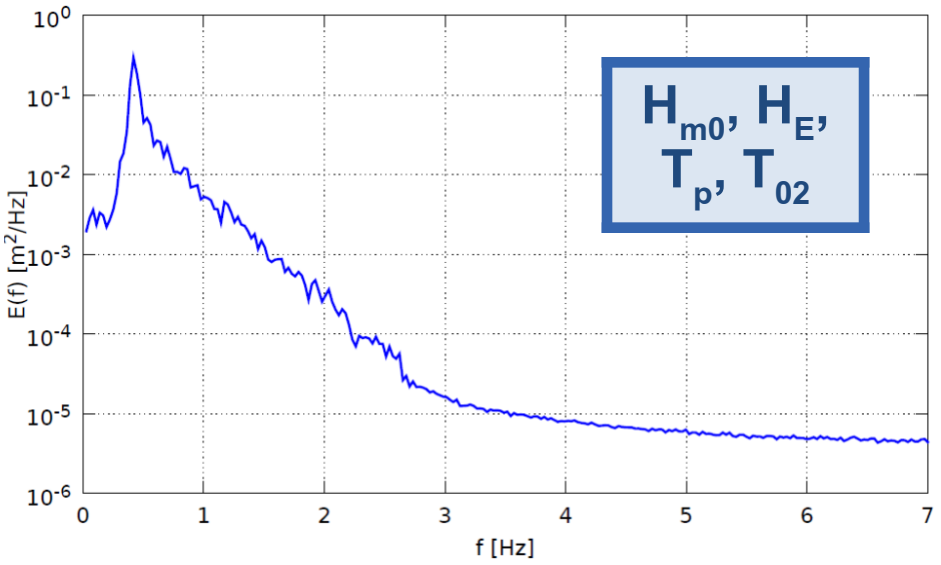
$\lambda_4 = 3$

Spectral or statistical approach

1. Statistical approach:
Wave height distribution
(wave-by-wave model)

2. Spectral approach:
Wave energy spectra
(spectral wave model)

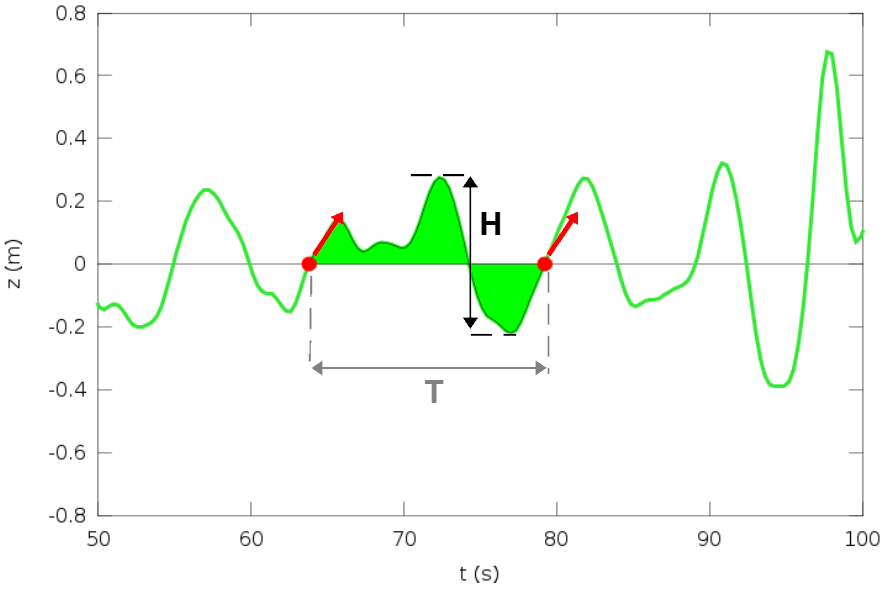




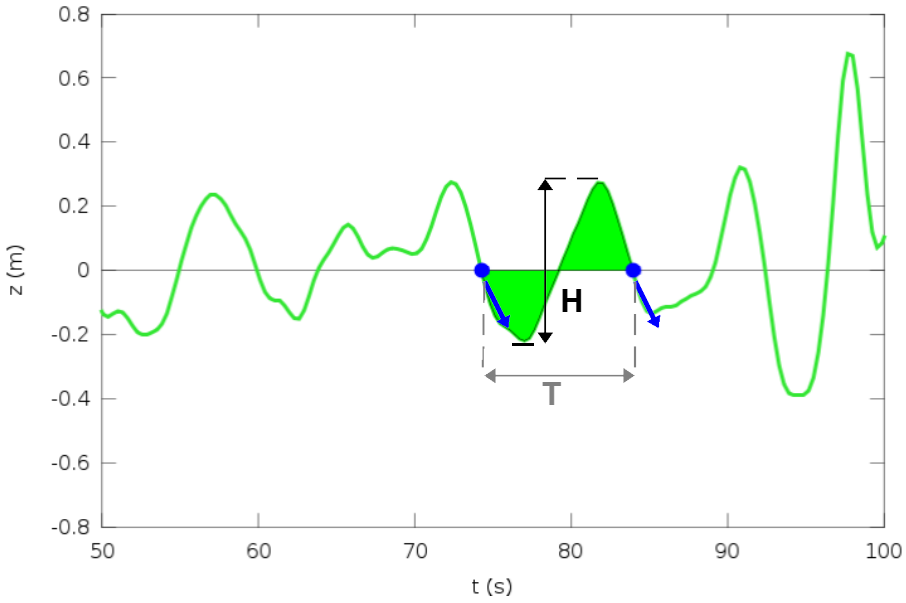
Statistical analysis

1. Identifying waves: zero-crossing method (wave-by-wave)

Upcrossing



Downcrossing



For each wave i in the time series (for i = 1, 2, … N), calculate (H_i, T_i)

Statistical analysis

2. Calculate the characteristic wave heights and periods

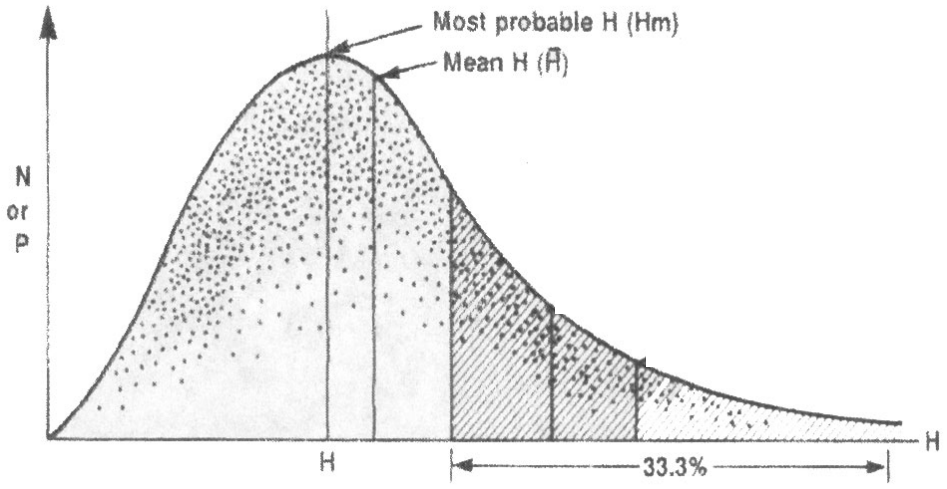
- Mean wave height and period:

$$\bar{H} = H_{moy} = H_m = \frac{1}{N}\sum_{i=1}^{N} H_i \qquad \bar{T} = T_{moy} = T_m = \frac{1}{N}\sum_{i=1}^{N} T_i$$

- Root-mean-square (rms) wave height:

$$H_{rms} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} H_i^2}$$

Wave height histogram or distribution



(from Bretschneider, 1964)

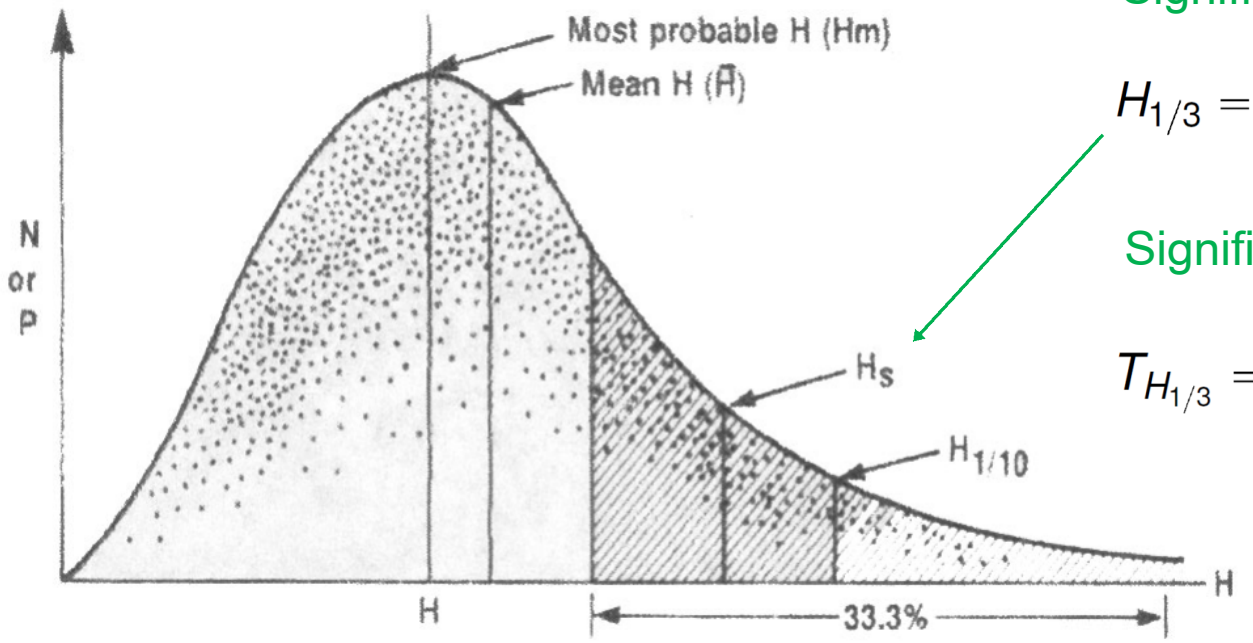
Statistical analysis

2. Calculate the characteristic wave heights and periods

- Sort (H,T) by **decreasing wave height:**

$$H_1(H_{max}) \geqslant H_2 \geqslant \ldots \geqslant H_N(H_{min})$$

- Calculate the mean wave height by fraction of largest waves and calculate the **corresponding wave period**:



Significant wave height

$$H_{1/3} = \boxed{H_s} = \frac{1}{N/3}\sum_{j=1}^{N/3} H_j$$

Significant wave period

$$T_{H_{1/3}} = \boxed{T_s} = \frac{1}{N/3}\sum_{j=1}^{N/3} T_j$$

Statistical analysis

2. Calculate the characteristic wave heights and periods

Mean of the 1/Q largest wave heights

$$H_{1/Q} = \frac{1}{N/Q} \sum_{j=1}^{N/Q} H_j$$

Mean of the 1/10 largest waves

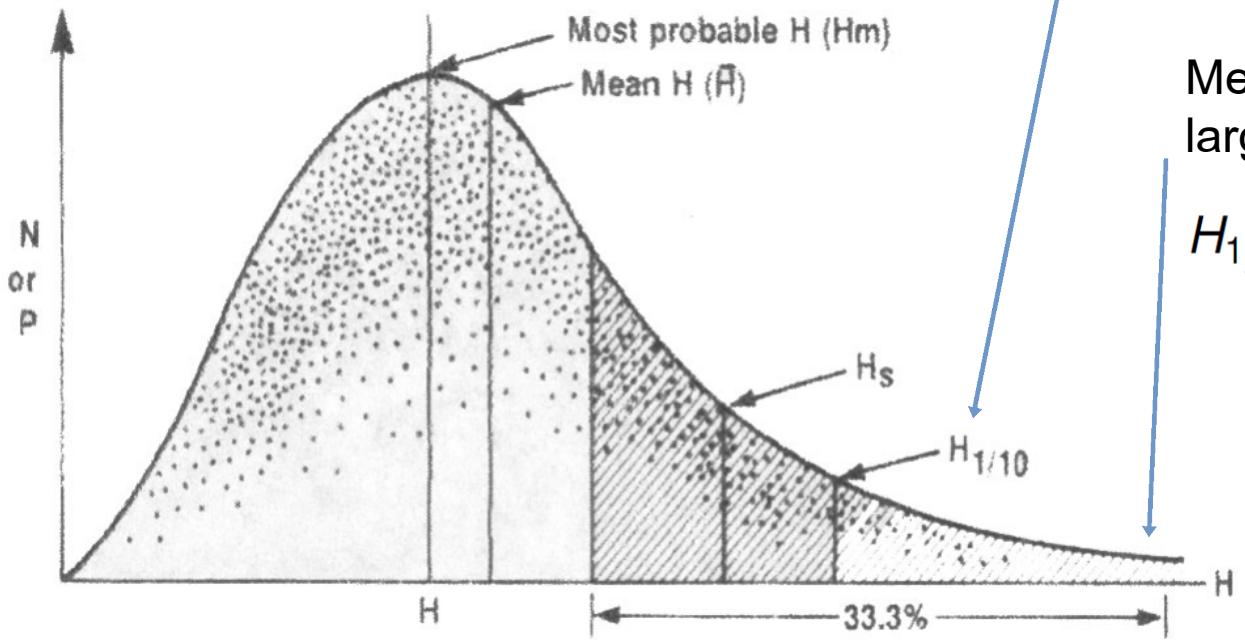
$$H_{1/10} = \frac{1}{N/10} \sum_{j=1}^{N/10} H_j$$

Mean of the 1/100 largest waves

$$H_{1/100} = \frac{1}{N/100} \sum_{j=1}^{N/100} H_j$$

Maximum recorded wave height

$$H_{max}$$



Statistical analysis

Summary of different wave heights

Characteristic wave parameter	Definition
Mean wave height H_m	$H_m = \frac{1}{N}\sum_{j=1}^{N} H_j$
Root-mean-square wave height H_{rms}	$H_{rms} = \sqrt{\frac{1}{N}\sum_{j=1}^{N} H_j^2}$
Significant wave height $H_s = H_{1/3}$	Average of the highest 1/3 of wave heights in the record
Wave height $H_{1/10}$	Average of the highest 1/10 of wave heights in the record
Wave height $H_{1/100}$	Average of the highest 1/100 of wave heights in the record
Wave height $H_{1/250}$	Average of the highest 1/250 of wave heights in the record
Wave height $H_{2\%}$	Wave height exceeded by 2% of wave heights in the record
Maximum wave height H_{max}	Highest wave height in the record

Statistical analysis

Summary of different wave periods

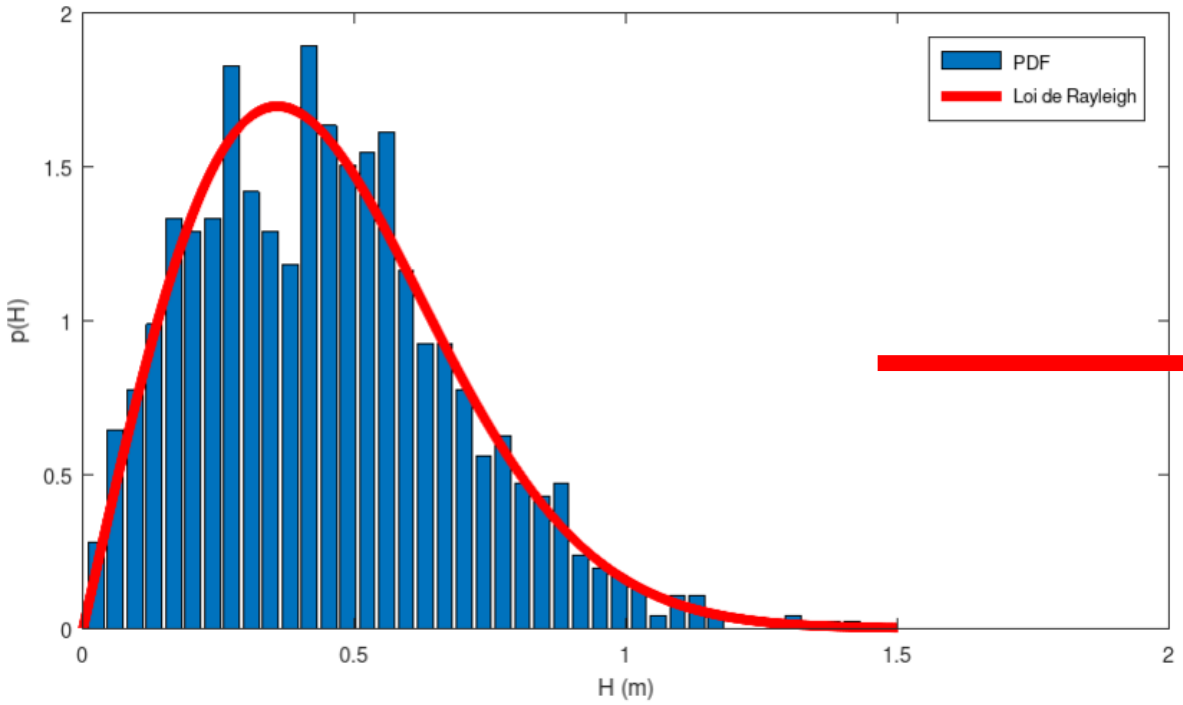
Characteristic wave parameter	Definition
Mean wave period T_m	$T_m = \frac{1}{N}\sum_{j=1}^{N} T_j$
Significant wave period $T_s = T_{1/3}$ (sometimes also referred to as $T_{H1/3}$)	Average of the periods associated with the largest 1/3 of wave heights in the record (ie average of periods of the waves selected to compute $H_{1/3}$)
Period of maximum wave height T_{Hmax}	Period of the largest wave height in the record
Maximum wave period T_{max}	Largest wave period in the record

Most commonly used values:
- wave height: H_s ($H_{1/3}$) et H_{rms}
- wave period: T_m et T_s

Typical length of wave time series used: 15-30 min

Statistical analysis

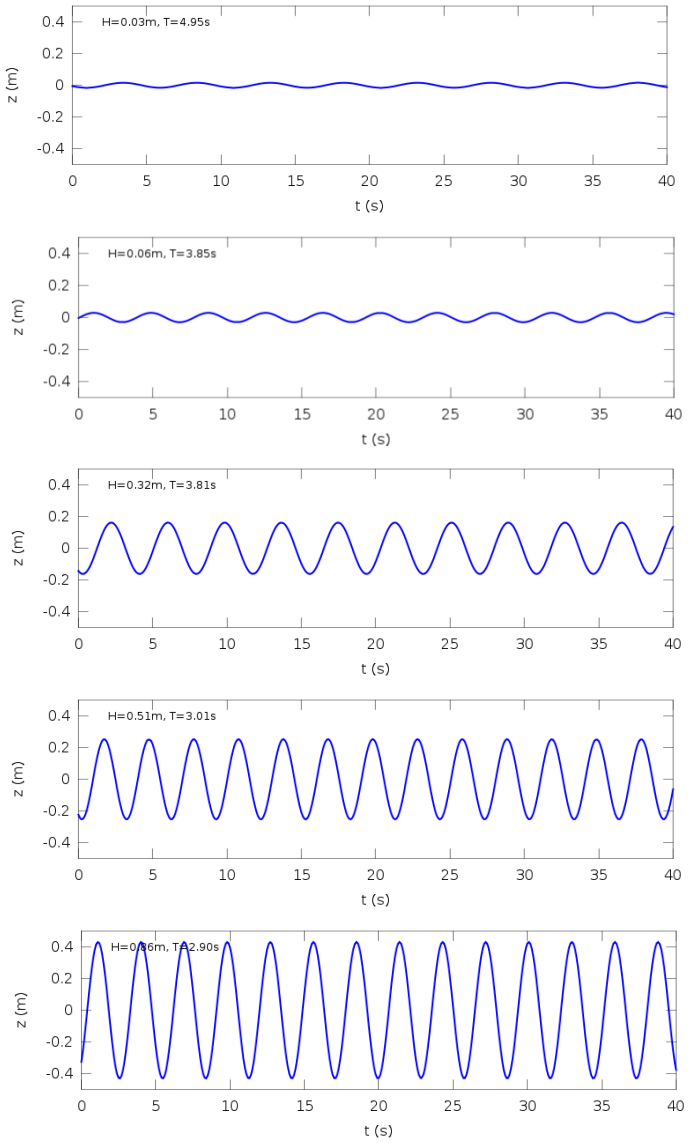
3. Wave height distribution laws



How can the wave height distribution be described ?

Sea state model

Superposition of sinusoidal waves



- Sinusoidal waves (H,T) :

$$\eta(t) = a\cos(2\pi ft - \Phi)$$

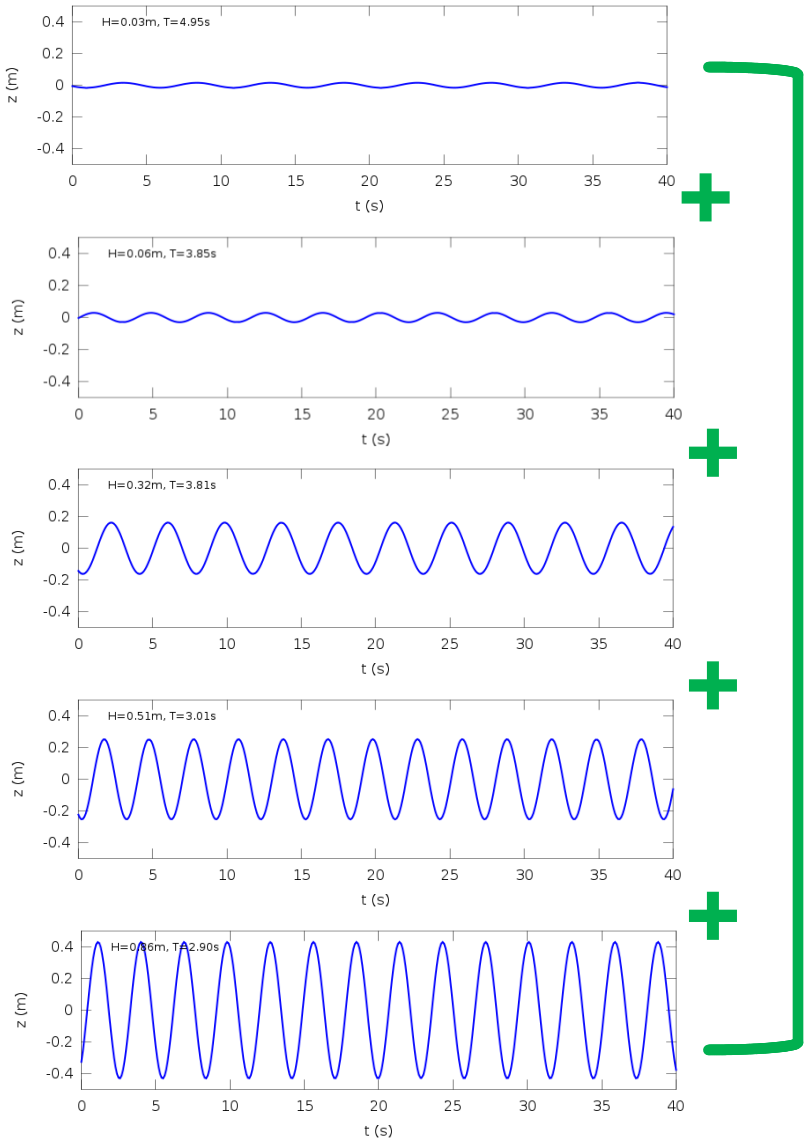
  - a = amplitude = H/2 (m)
  - f = frequency= 1/T (Hz)
  - φ = phase (rad)

- Sum of sinusoidal waves:

$$\eta(t) = \sum_{i=1}^{N} \eta_i(t) = \sum_{i=1}^{N} a_i \cos(2\pi f_i t - \Phi_i)$$

Sea state model

Superposition of sinusoidal waves



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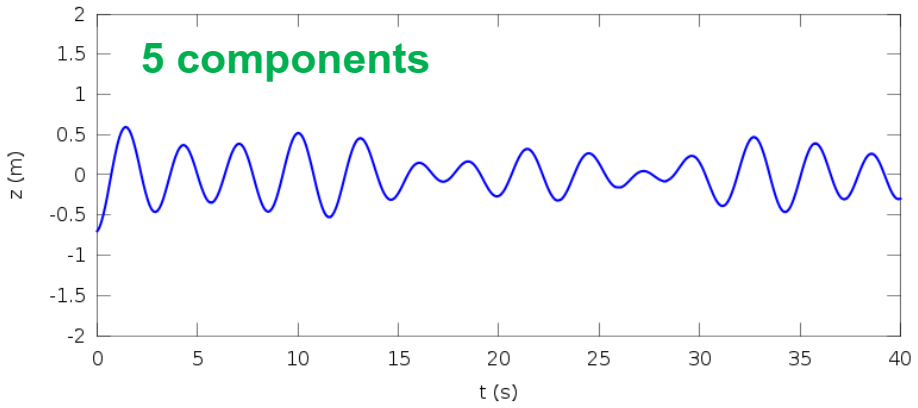
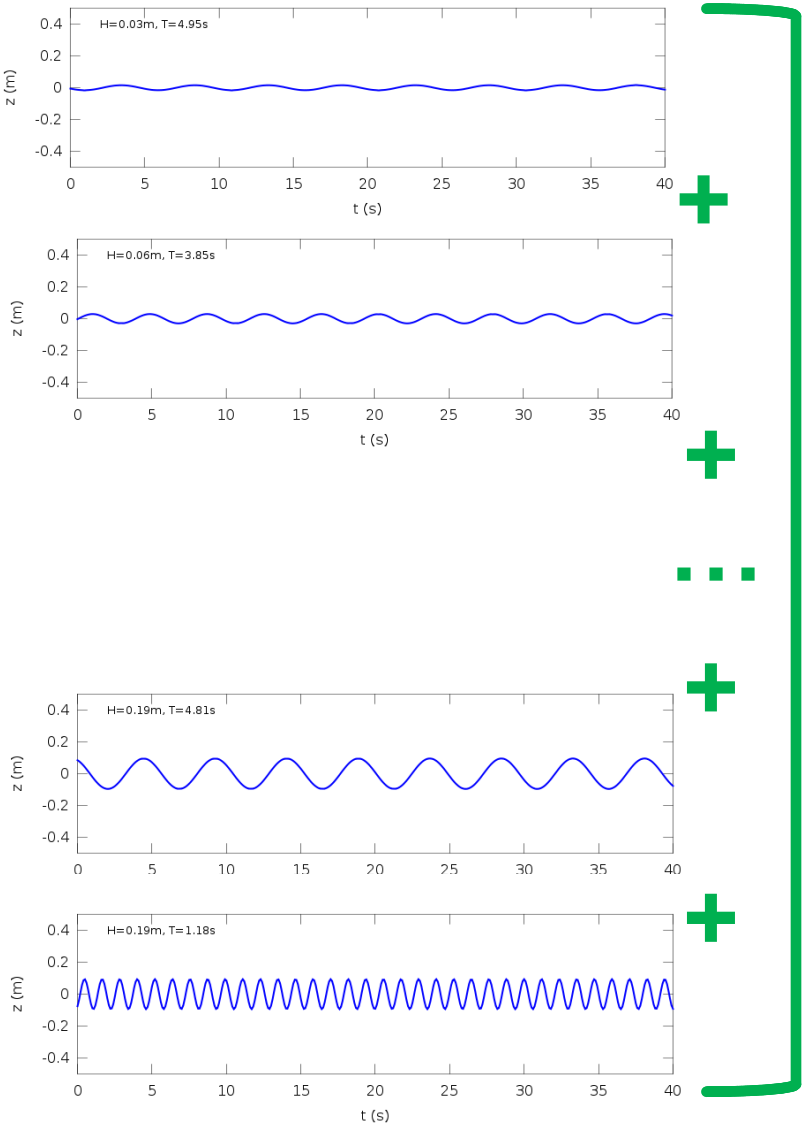


Sea state model

Superposition of sinusoidal waves



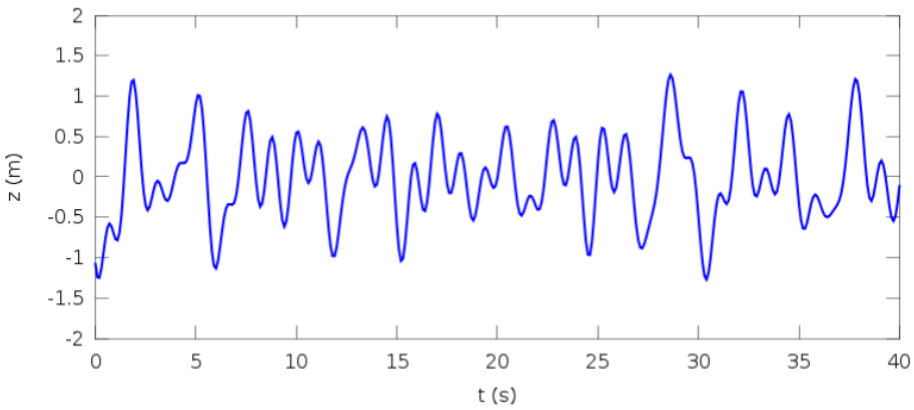
- Sinusoidal waves (H,T) :

$$\eta(t) = a\cos(2\pi ft - \Phi)$$

  - a = amplitude = H/2 (m)
  - f = frequency= 1/T (Hz)
  - φ = phase (rad),
    (also represented by α)

- Sum of sinusoidal waves:

20 components



Sea state model

Represent the free surface with a random-phase/amplitude model

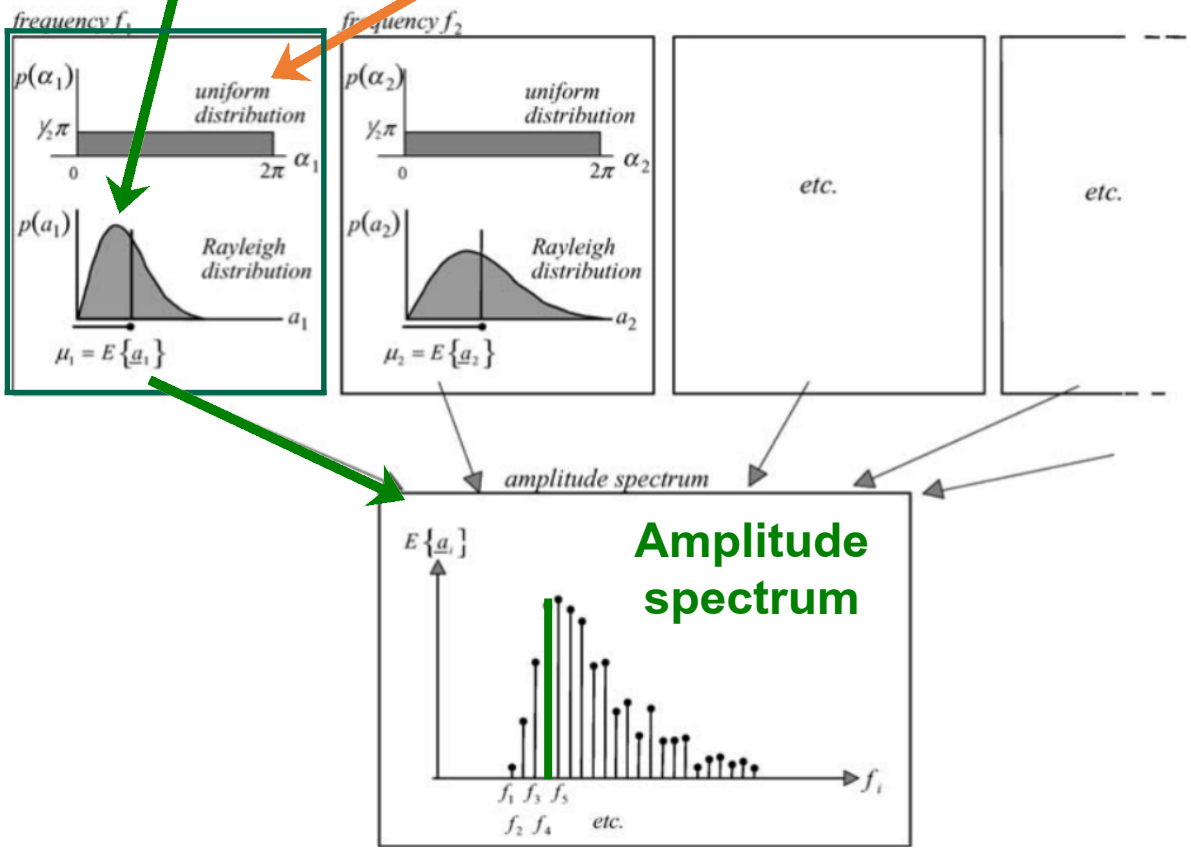
For each frequency f_i, ...

$$\underline{\eta}(t) = \sum_{i=1}^{N} \underline{a}_i \cos(2\pi f_i t + \underline{\alpha}_i)$$

random variables

(1) represent the **phase (uniform)** and **amplitude (Rayleigh)** distribtions

(2) calculate the **expected value of the amplitude**

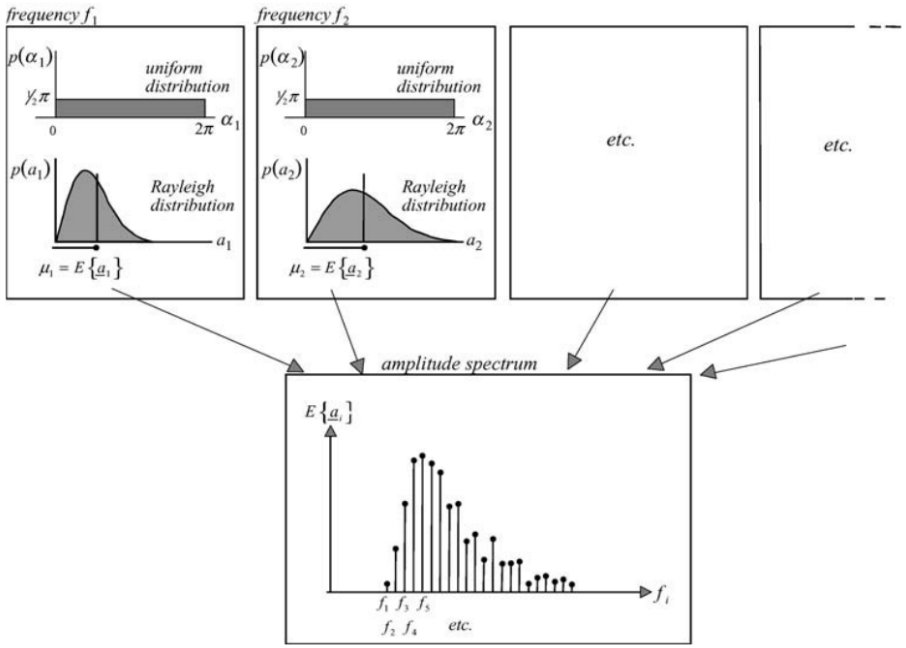


Sea state model

Amplitude distribution

$$p(\eta) = \frac{1}{(2\pi m_0)^{1/2}} \exp\left(-\frac{\eta^2}{2m_0}\right)$$

$$E[\eta] = 0 \qquad E\left[\eta^2\right] = m_0$$



Assumption: stationary and Gaussian processes with random variables:

- phase (uniform distribution)
- amplitude (Rayleigh distribution)

$$p(a_i) = \frac{\pi}{2}\frac{a_i}{\mu_i^2} \exp\left(-\frac{\pi a_i^2}{4\mu_i^2}\right), \text{ with } \mu_i = E[a_i]$$

(Holthuijsen, 2007)

Sea state model

Gaussian sea state model → Rayleigh distribution

Evaluate these assumptions at sea:

- **Stationarity**: break the wave time series up into 15 to 30-minute segments

- **Independent:** wave components interact, but if waves are:
  - NOT too steep and
  - NOT in shallow water

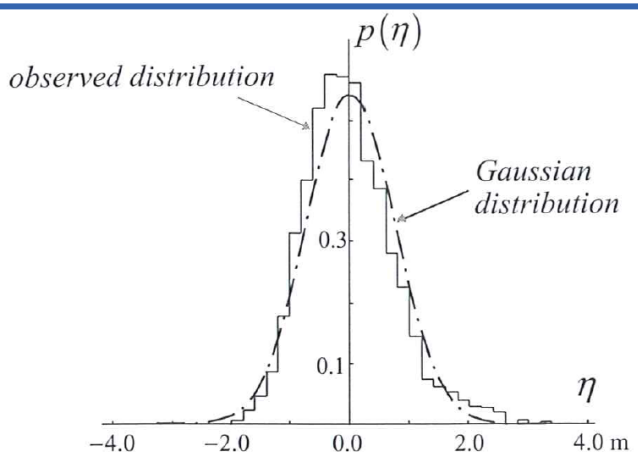
  we can consider these effects to be weak

(Holthuijsen, 2007)

Sea state model

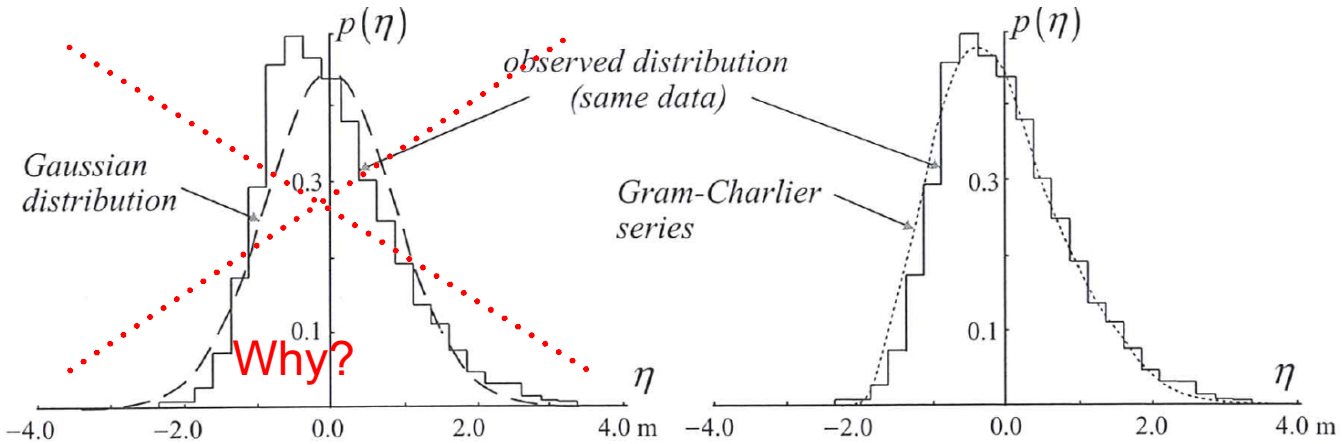
Assumption of a Gaussian model

***STEEP* waves in deep water**



Gaussian model

***LARGE* waves in shallow water**



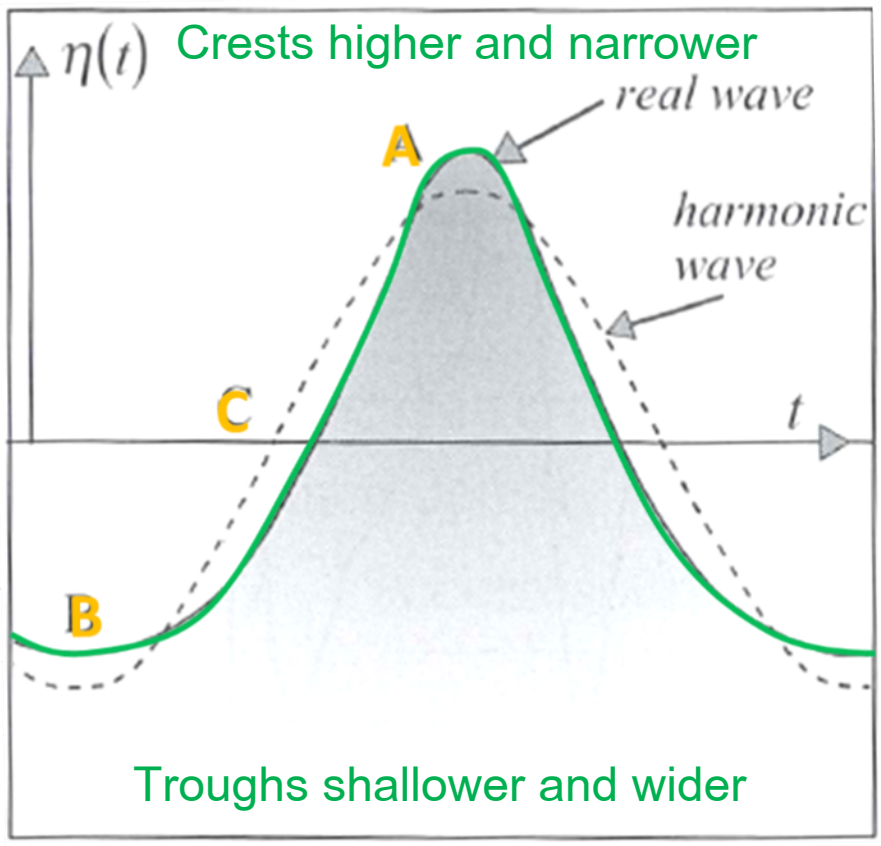
Gaussian model → Non-Gaussian model

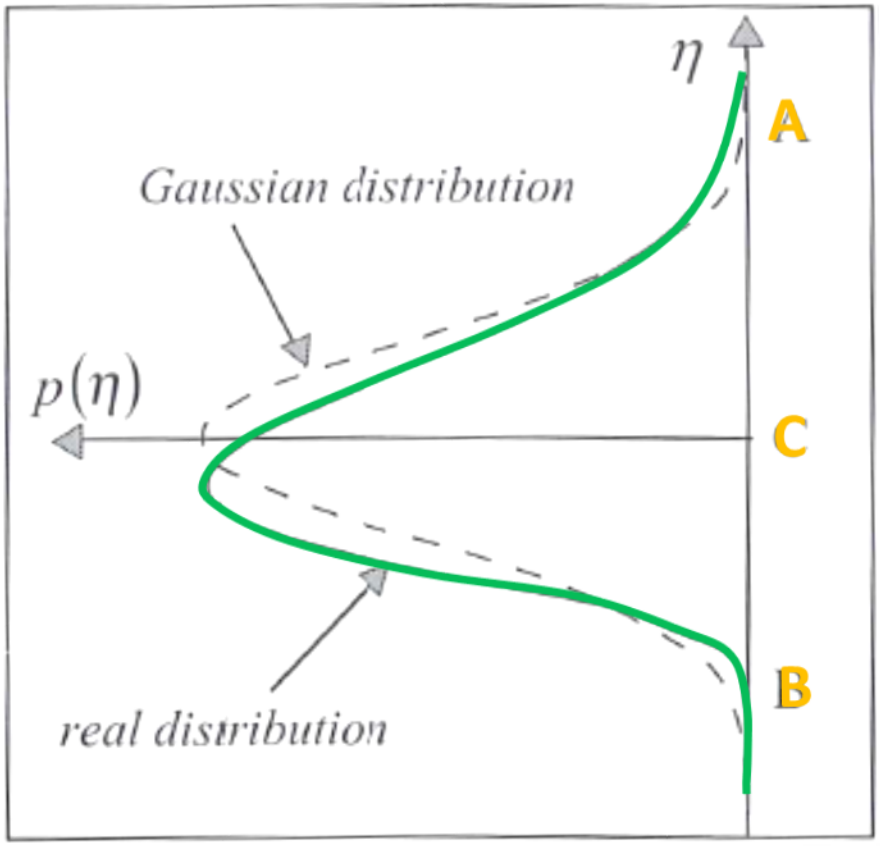
(Holthuijsen, 2007)

Sea state model

Assumption of a Gaussian model

Nonlinear waves





Other distributions are more well adapted for nonlinear waves (e.g. Gram-Charlier)

(Holthuijsen, 2007)

Sea state model

Rayleigh distribution

Valid in deep and intermediate water for waves that are not too steep:

- Probability density function:

$$p(H) = \frac{\pi}{2}\frac{H}{H_m^2}\exp\left(-\frac{\pi}{4}\left(\frac{H}{H_m}\right)^2\right) = \frac{2H}{H_{rms}^2}\exp\left(-\left(\frac{H}{H_{rms}}\right)\right)^2$$

- Cumulative distribution function:

$$P(H) = 1 - \exp\left(-\frac{\pi}{4}\left(\frac{H}{H_m}\right)^2\right) = 1 - \exp\left(-\left(\frac{H}{H_{rms}}\right)^2\right)$$

Requires only one wave parameter (H_m, m_0, H_{rms}) to calculate other wave statistics:

- $H_{P\%}$ (wave height exceeded by P% of waves)

$$\frac{H_{P\%}}{H_{rms}} = \sqrt{-ln\left(\frac{P}{100}\right)}$$

- $H_{1/Q}$ (mean of 1/Q of the largest waves)

$$\frac{H_{1/Q}}{H_{rms}} = Q\,erfc\left(\sqrt{lnQ}\right) + \sqrt{lnQ} \qquad erfc(x) = \int_x^{+\infty}\exp\left(-t^2\right)dt$$

Sea state model

Rayleigh distribution

Rayleigh distribution of the wave heights

Calculation of $H_{1/3} = H_s$: 2 steps

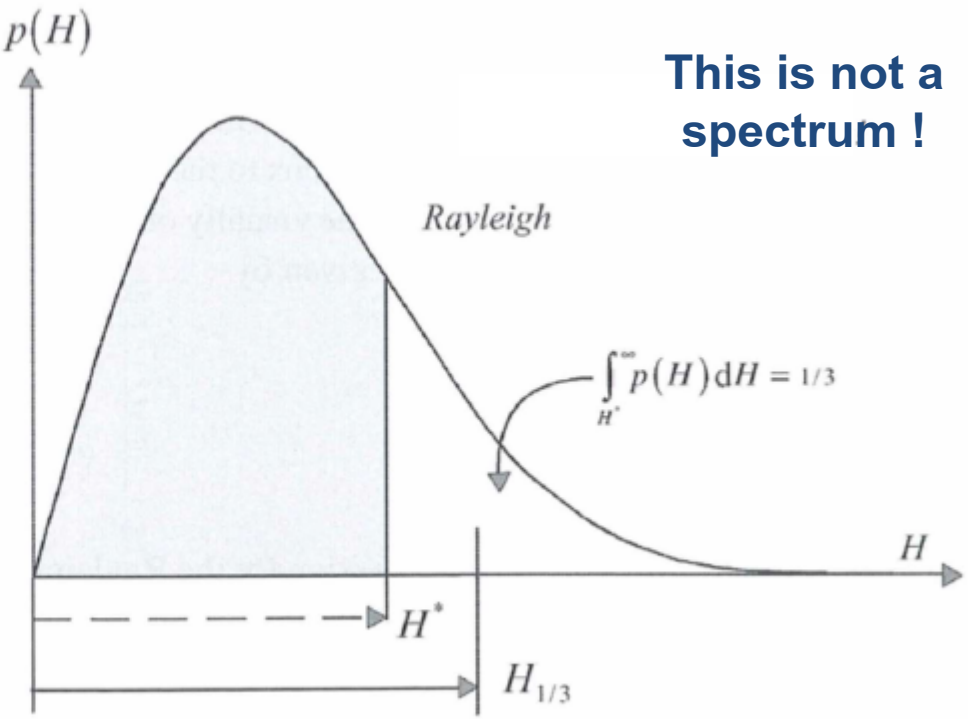
1. Determine $H^* = H_{33.33\%}$ = height exceeded by 33.33 % (= 1/3) of the (individual) waves:

$$\int_{H^*}^{\infty} p(H)\mathrm{d}H = \frac{1}{3}$$

$$\Rightarrow \quad H^* = \sqrt{-\ln 3}\, H_{rms} \approx 1.048 H_{rms}$$

2. Compute the mean of heights lying between H^* and $+\infty$.

$$H_{1/3} = \frac{\int_{H^*}^{\infty} H\ p(H)dH}{\int_{H^*}^{\infty} p(H)dH} = 3\int_{H^*}^{\infty} H\ p(H)dH$$



This is not a spectrum !

One finds: $H_{1/3} \approx 1.6\ H_m \approx 1.416\ H_{rms} \approx 4.004\ m_0^{1/2}$ => $H_{1/3} \approx 4\ m_0^{1/2}$

(Holthuijsen, 2007)

Sea state model

Rayleigh distribution

Characteristic height H	Wave height ratios			
	$H/\sqrt{m_0}$	H/H_m	H/H_{rms}	H/H_s
Standard deviation of free surface $\sigma_\eta = \sqrt{m_0}$	1	0.399	0.353	0.250
Mean wave height H_m	2.507	1	0.886	0.626
Root-mean-square wave height H_{rms}	2.828	1.128	1	0.706
Significant wave height $H_s = H_{1/3}$	4.004	1.597	1.416	1
Wave height $H_{1/10}$	5.090	2.031	1.800	1.273
Wave height $H_{1/100}$	6.673	2.662	2.359	1.668
Wave height $H_{2\%}$	5.594	2.232	1.978	1.397

H_{max} = ?
largest

Sea state model

Rayleigh distribution

- Rayleigh distribution is *unbounded* : every wave height has a non-zero probability
  - Cannot calculate H_{max} in a deterministic way
  - Statistical distribution of r.v. H_{max}/Hs can be estimated:

Mode of the variable H_{max}/H_s ***:***
(= most probable value)

$$\left[\frac{H_{max}}{H_s}\right]_{mode} \approx \sqrt{\frac{\ln N}{2}}$$

Mean of the variable H_{max}/H_s ***:***

$$\left[\frac{H_{max}}{H_s}\right]_{mean} \approx \left(\sqrt{\frac{\ln N}{2}} + \frac{\gamma}{2\sqrt{2\ln N}}\right)$$

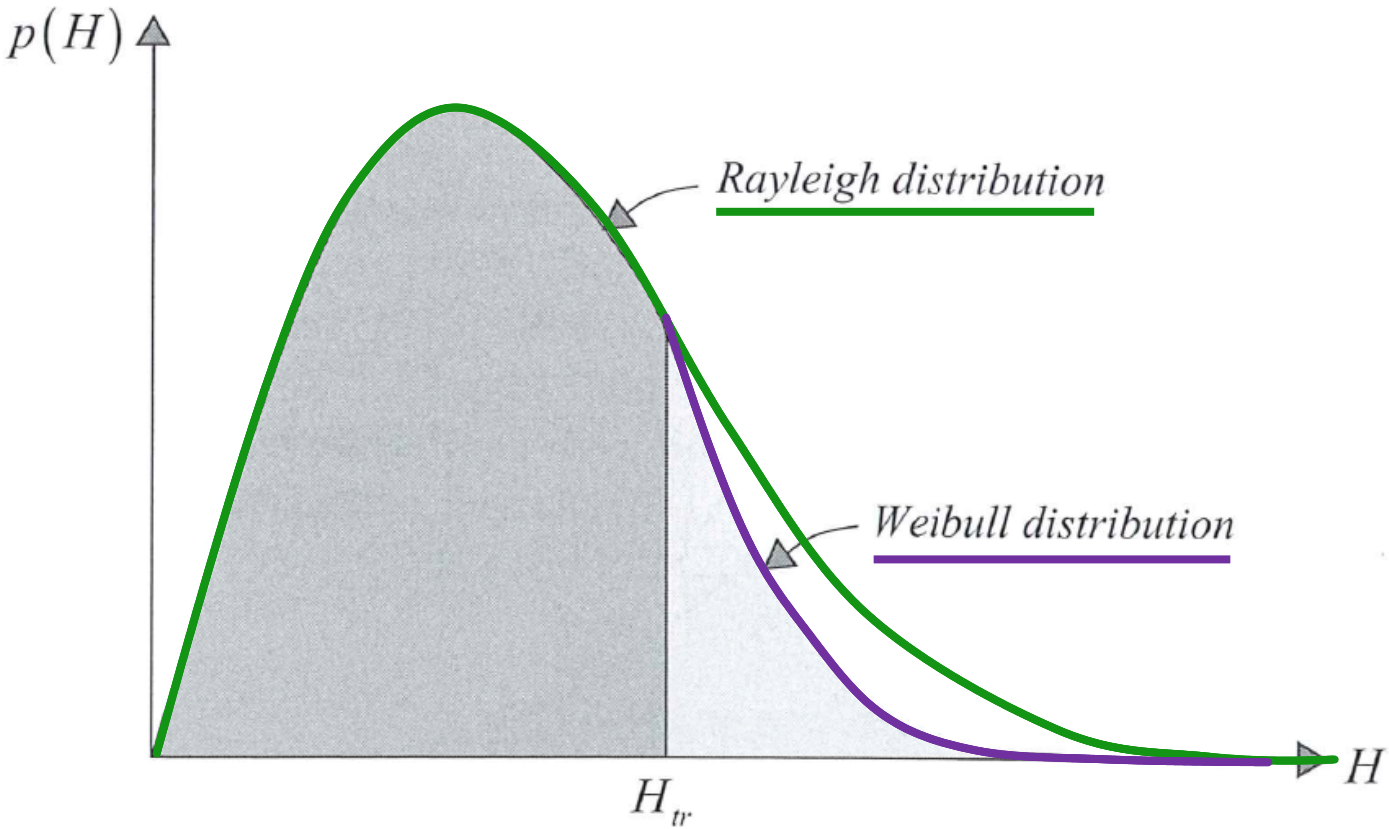
$\gamma =$ Euler constant ≈ 0.5772.

Wave height distribution property	Number of waves, *N*, for the mode and mean values						
	100	200	500	1000	2000	5000	10 000
$(H_{max}/H_s)_{mode}$	1.52	1.63	1.76	1.86	1.95	2.06	2.15
$(H_{max}/H_s)_{mean}$	1.61	1.72	1.84	1.94	2.02	2.13	2.21

Sea state model

Shallow water distributions:

- Beta-Rayleigh (Hughes and Borgman, 1987)
- Weibull (Glukhovskiy, 1966)
- Mendez et al (2004), modified
- Weibull composite (CWD) (Battjes et Groenendijk, 2000)



(Holthuijsen, 2007)

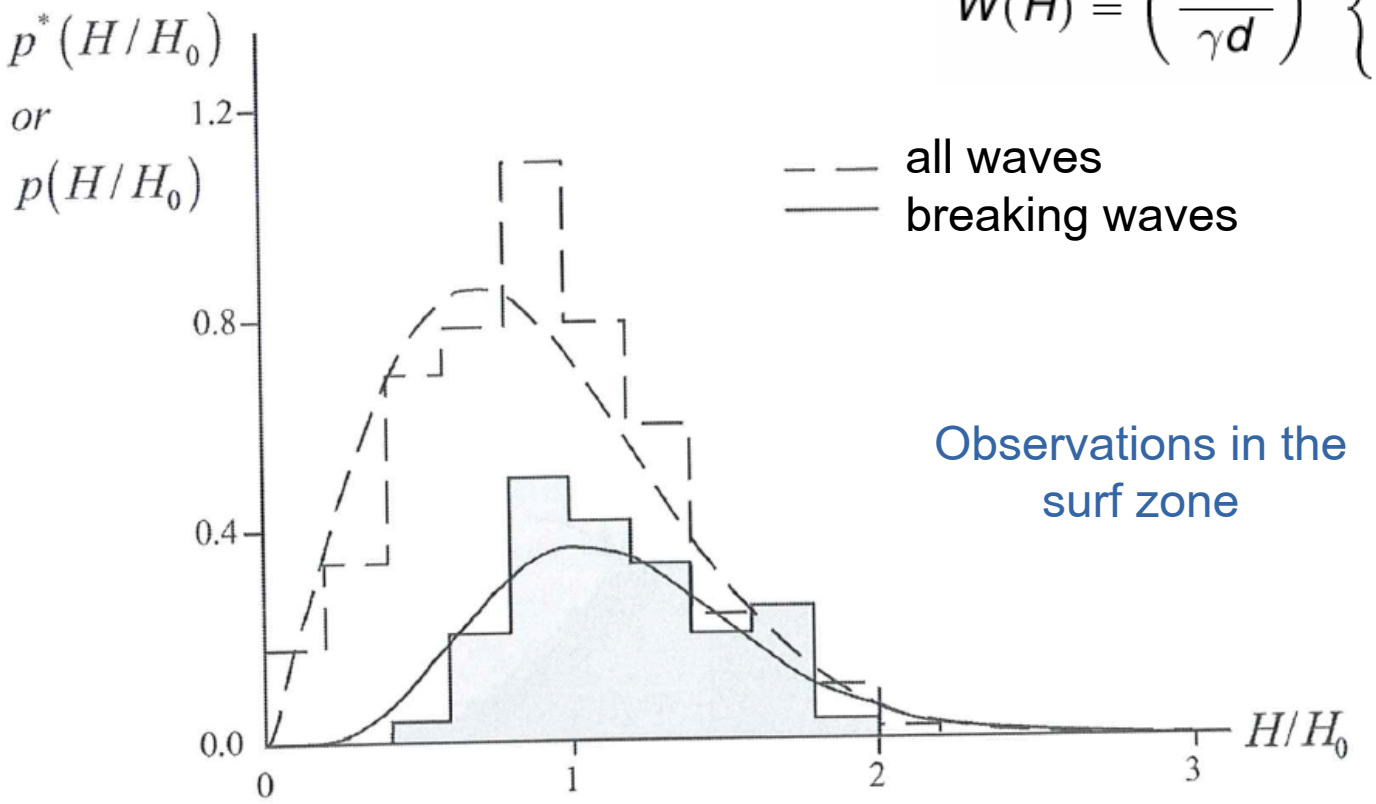
Sea state model

Shallow water distribution with breaking waves (Thornton and Guza, 1983):

$$p*(H_{br}) = \frac{2H}{H_{rms}^2}\exp\left(-\left(\frac{H}{H_{rms}}\right)\right)^2 W(H)$$

$$W(H) = \left(\frac{H_{rms}}{\gamma d}\right)^n \left\{1 - \exp\left[-\left(\frac{H}{\gamma d}\right)^2\right]\right\}$$

γ=0.42, n=4



Observations in the surf zone

(Holthuijsen, 2007)

Sea state model

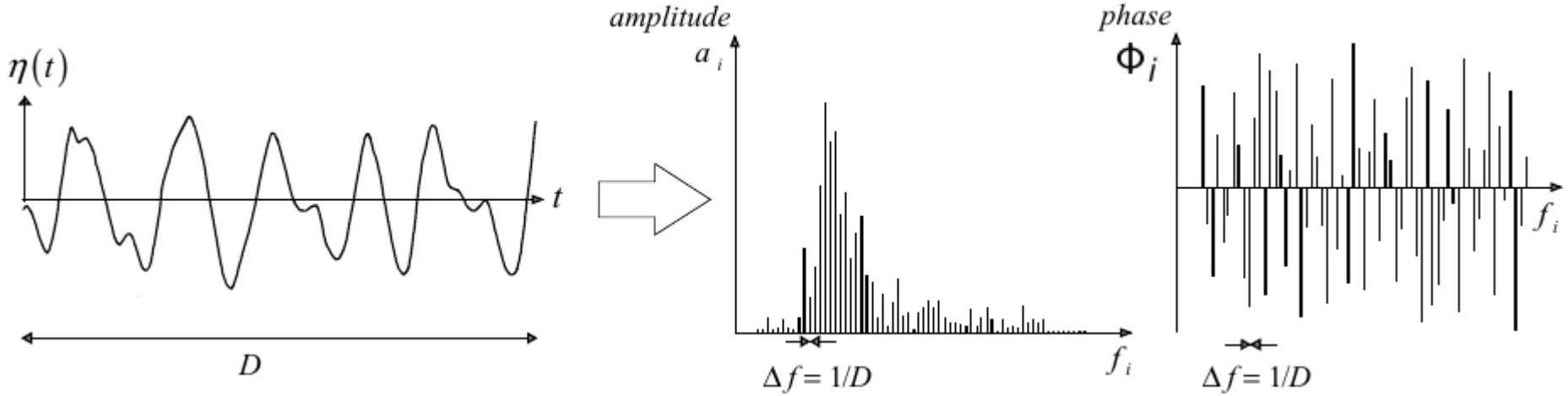
Represent the free surface with a random-phase/amplitude model

$$\eta(t) = \sum_{i=1}^{N} \eta_i(t) = \sum_{i=1}^{N} a_i \cos(2\pi f_i t - \Phi_i)$$

$$f_i = i\Delta f \qquad \Delta f = 1/D$$

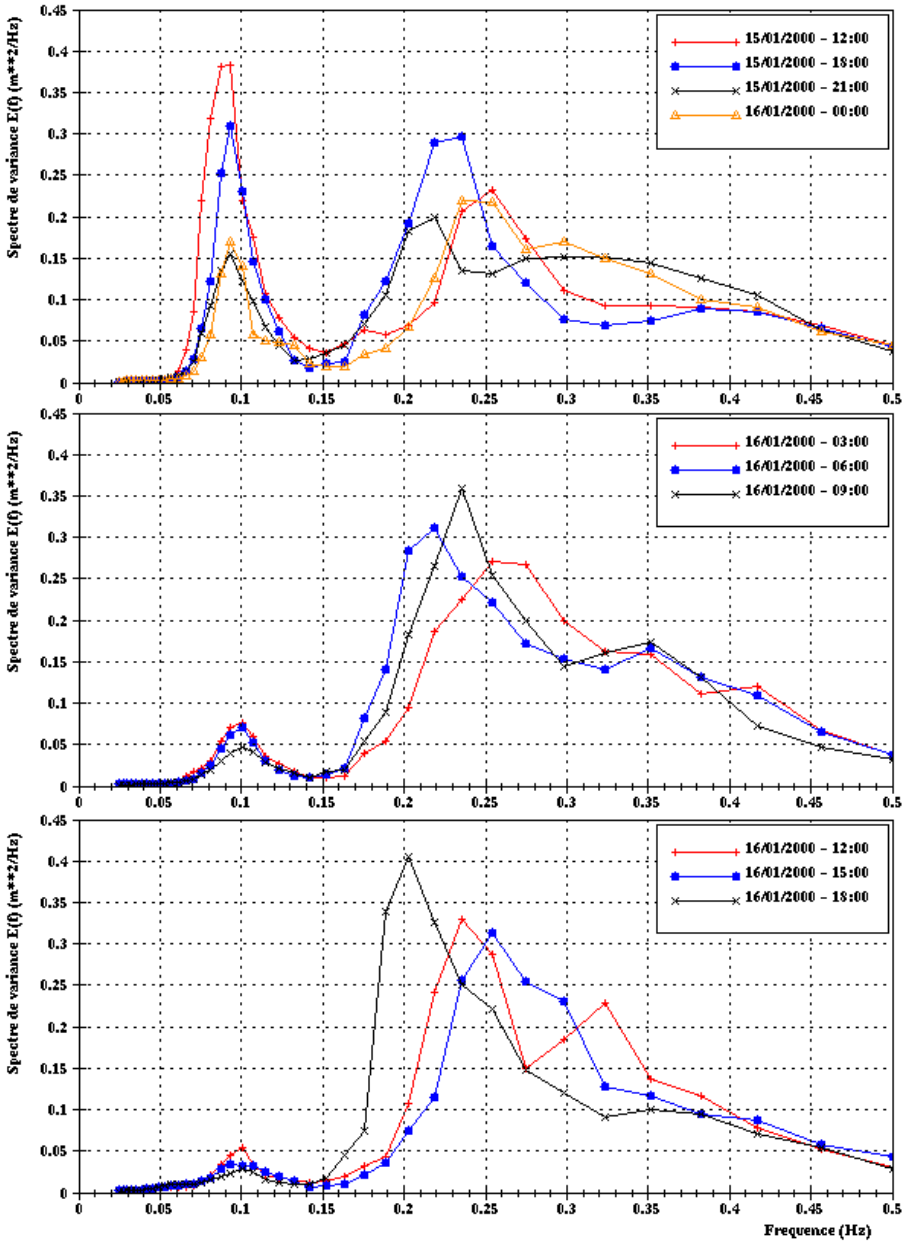
D= length of time series

Amplitude spectrum **Phase spectrum**



Uncorrelated phases Φ_i , independents components η_i

Spectral analysis

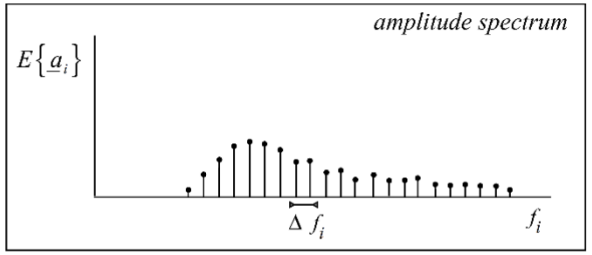


Datawell Waverider Buoy
Cerema
Ile d'Yeu
15-16/01/2000

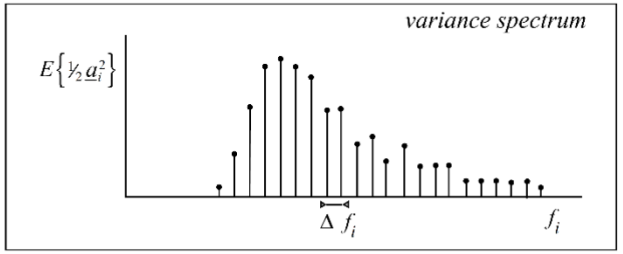
Spectral analysis

Decomposing the time series in frequency components

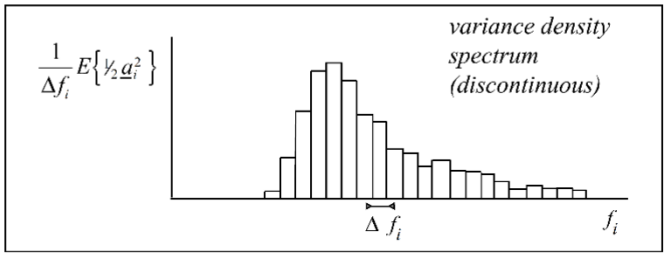
Amplitude spectrum



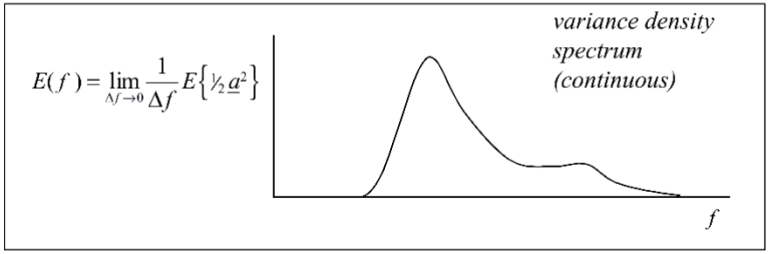
Discrete variance spectrum



Discontinuous variance density spectrum



Continuous variance density spectrum



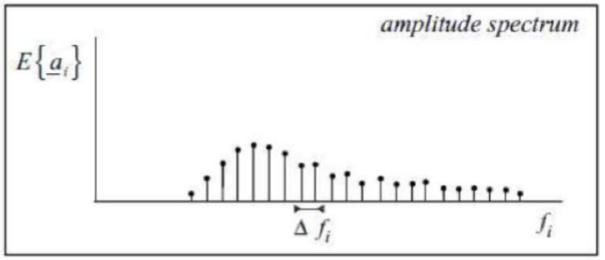
(Holthuijsen, 2007)

Spectral analysis

Decomposing the time series in frequency components

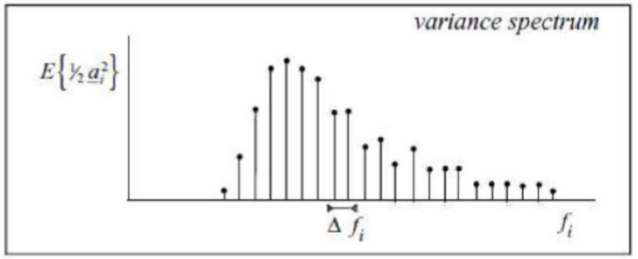
Amplitude spectrum

$$\eta(t) = \sum_{i=1}^{N} \underline{a}_i \cos\left(2\pi f_i t + \underline{\alpha}_i\right)$$



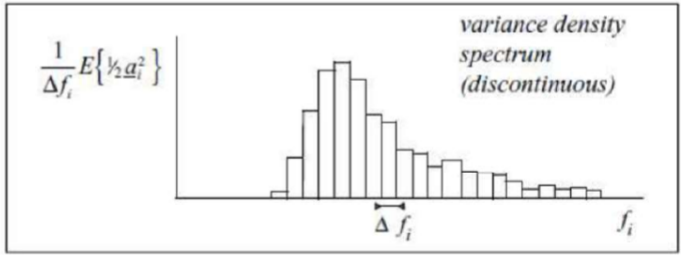
Discrete variance spectrum

$$\hat{E}(f_i) = \mathrm{E}\left\{\frac{1}{2}\underline{a}_i^2\right\} \qquad m_0 = \sum_{i=1}^{N} \hat{E}(f_i)$$



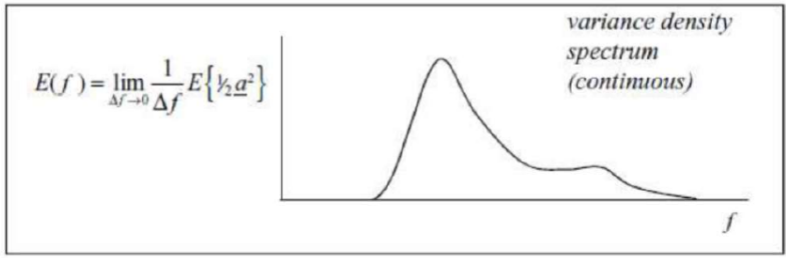
Discontinuous variance density spectrum

$$E^*(f_i) = \frac{1}{\Delta f_i} E\left\{\frac{1}{2}\underline{a}_i^2\right\} \qquad m_0 = \sum_{i=1}^{N} E^*(f_i).\Delta f_i$$



Continuous variance density spectrum

$$E(f) = \lim_{\Delta f \to 0} \frac{1}{\Delta f} E\left\{\frac{1}{2}\underline{a}^2\right\} \qquad m_0 = \int E(f)\, df$$



(Holthuijsen, 2007)

Spectral analysis

Decomposing the time series in frequency components

Powerful tool:

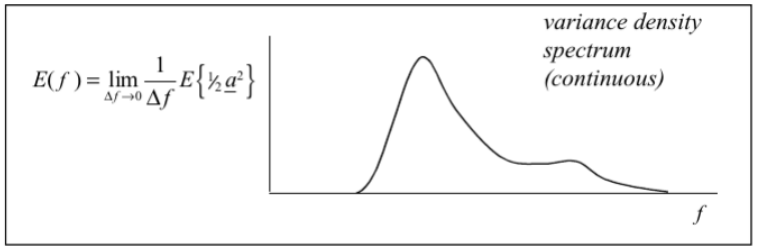
Statistical description of the surface elevation
(assuming that surface elevation is a stationary, Gaussian process)

Calculate statistical characteristics of the wave field

Continuous variance spectrum

$$E(f) = \lim_{\Delta f \to 0} \frac{1}{\Delta f} E\left\{\tfrac{1}{2}\underline{a}^2\right\}$$

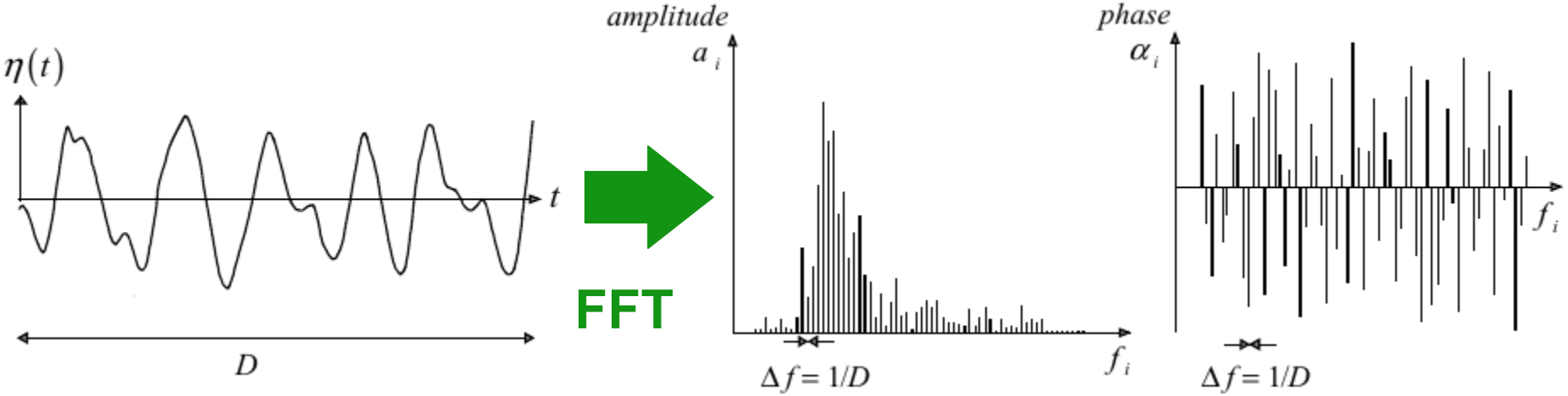
$$m_0 = \int E(f)\, df$$



Units: m^2/Hz

Spectral analysis

What does a wave spectra represent?



Synthetic presentation of the wave energy as a function of frequency:

- Dominant frequencies
- Distribution of energy as a function of frequency

(Holthuijsen, 2007)

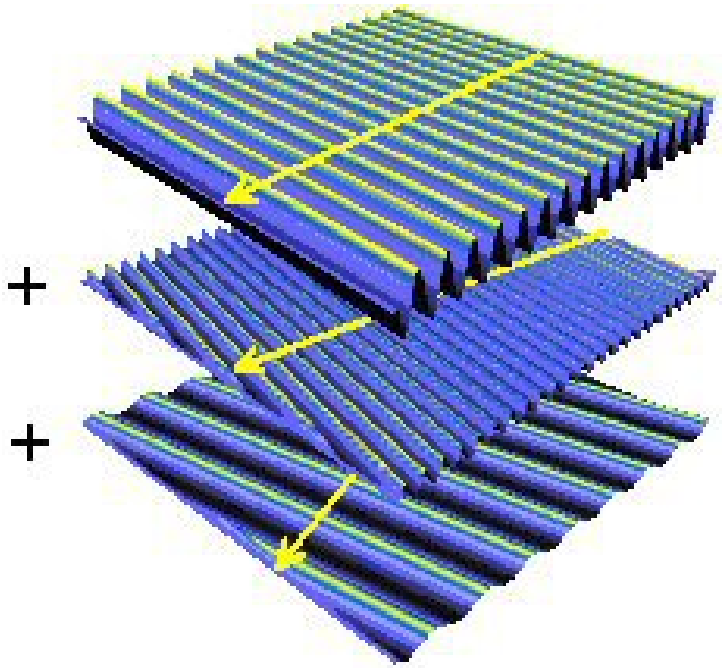
frequency = f, ω

Spectral analysis

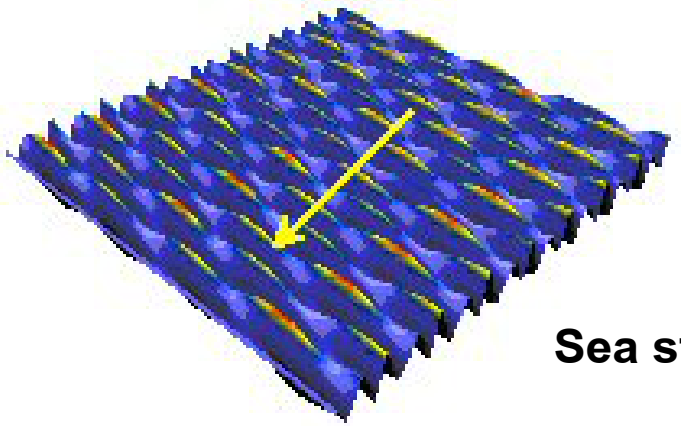
Sum of harmonic components

$$\eta(x,y,t) = \sum_{i=1}^{M}\sum_{j=1}^{N} a_{i,j}\cos\left(\omega_i t - k_i x\cos\theta_j - k_i y\sin\theta_j + \alpha_{i,j}\right)$$

Amplitude — frequency — Wave number — Wave angle — Phase



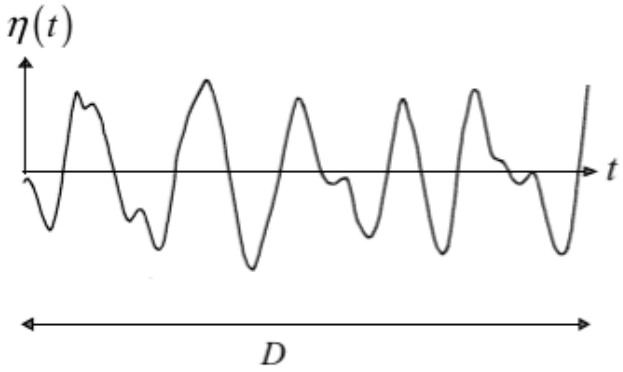
Sum of *i* frequencies
Sum of *j* directions



Sea state

Spectral analysis

What does this mean?



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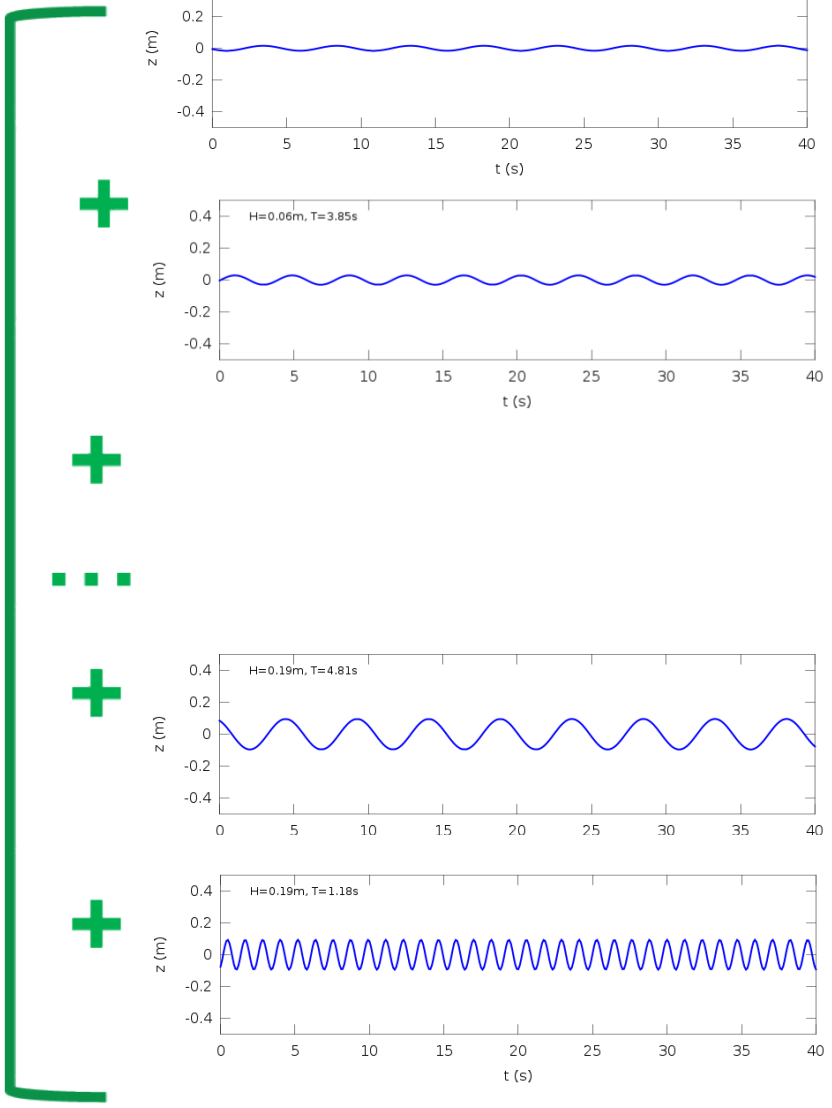
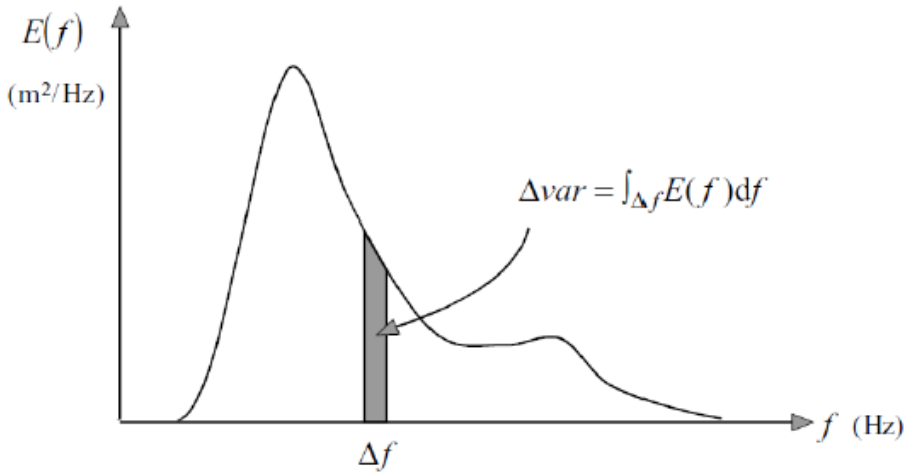


Remember: (Gaussian model)

Sea-state = sum of many sinusoidal wave components

Spectral analysis

What does this mean?



Contribution of variance from each frequency band

$$\Delta var = \int_{\Delta f} E(f)\mathrm{d}f$$

$$\overline{\eta^2} = \int_0^{\infty} E(f)\mathrm{d}f$$

Relationship between the energy and the variance: $E_{total} = \rho g \overline{\eta^2}$

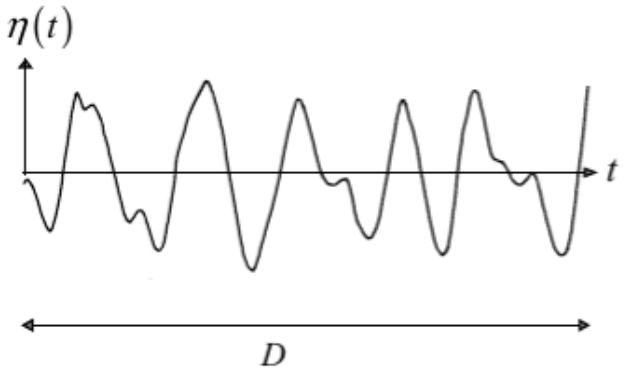
Energy density spectrum <=> Variance density spectrum

$$E_{energy}(f) = \rho g E_{variance}(f)$$

Units: J/Hz

Spectral analysis

Alternative mathematical definition of the spectrum:



Autocorrelation function:

$$R(\tau) = E\{\eta(t)\eta(t+\tau)\}$$

Power density spectrum:

$$G(\omega)=\int_{-\infty}^{+\infty} R(\tau)\cdot e^{-i\omega\tau}\,d\tau$$

Simplify by using one-sided function S (since G is even): $S(\omega)=2G(\omega)$

Wave spectrum (S)

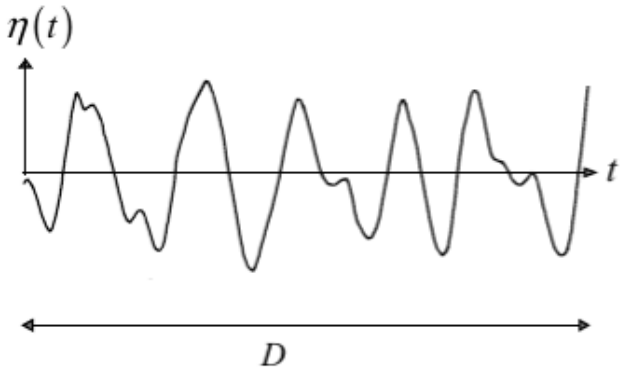
$$S(\omega)=\frac{2}{\pi}\int_{0}^{+\infty} R(\tau)\cdot\cos(\omega t)\,d\tau$$

Autocorrelation function (R)

$$R(\tau)=\int_{0}^{+\infty} S(\omega)\cdot\cos(\omega t)\,d\omega$$

Spectral analysis

What does this mean?



Gaussian model

$$\eta(t) = \sum A_i \cdot \cos(k_i x - \omega_i t - \varphi_i)$$

Thus, by definition, ….
the power density spectrum:

$$\eta(t) \cdot \eta(t+\tau) = \sum\sum A_i \cos(\omega_i t + \varphi_i) \cdot A_j \cos(\omega_j (t+\tau) + \varphi_j)$$

and autocorrelation function:

$$R(\tau) = E\{\eta(t)\eta(t+\tau)\} = \sum \frac{{A_i}^2}{2} \cdot \cos(\omega_i \tau)$$

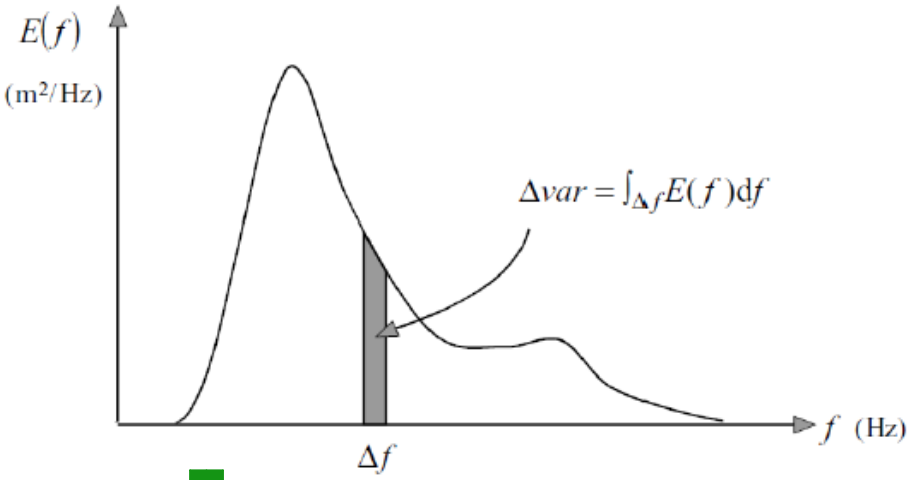
Spectral analysis

What does this mean?

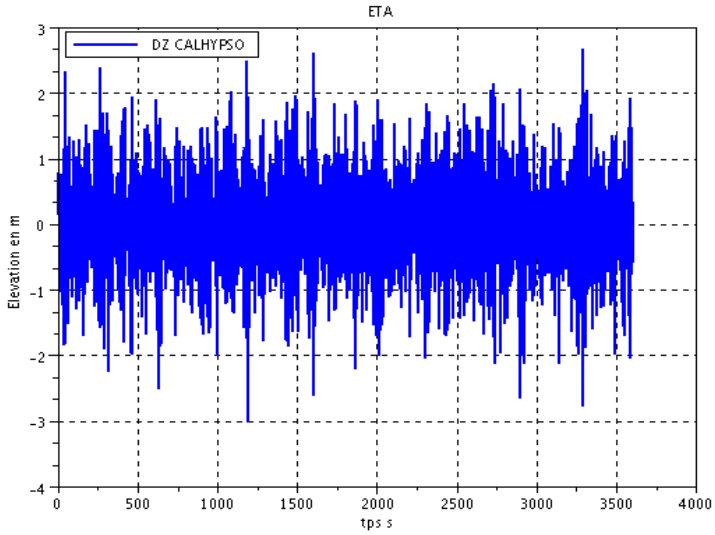
$$R(\tau) = \sum \frac{A_i^2}{2} \cdot \cos(\omega_i \tau)$$

$$R(\tau) = \int_0^{+\infty} S(\omega) \cdot \cos(\omega\tau)\, d\omega$$

$$S(\omega)\, d\omega = \frac{A_i^2}{2}$$



FFT



$$S(\omega)\, d\omega \rightarrow \frac{A_i^2}{2} = S(\omega)\, d\omega \rightarrow \eta(t) = \sum A_i \cdot \cos(\omega_i t - \varphi_i)$$

Spectral analysis

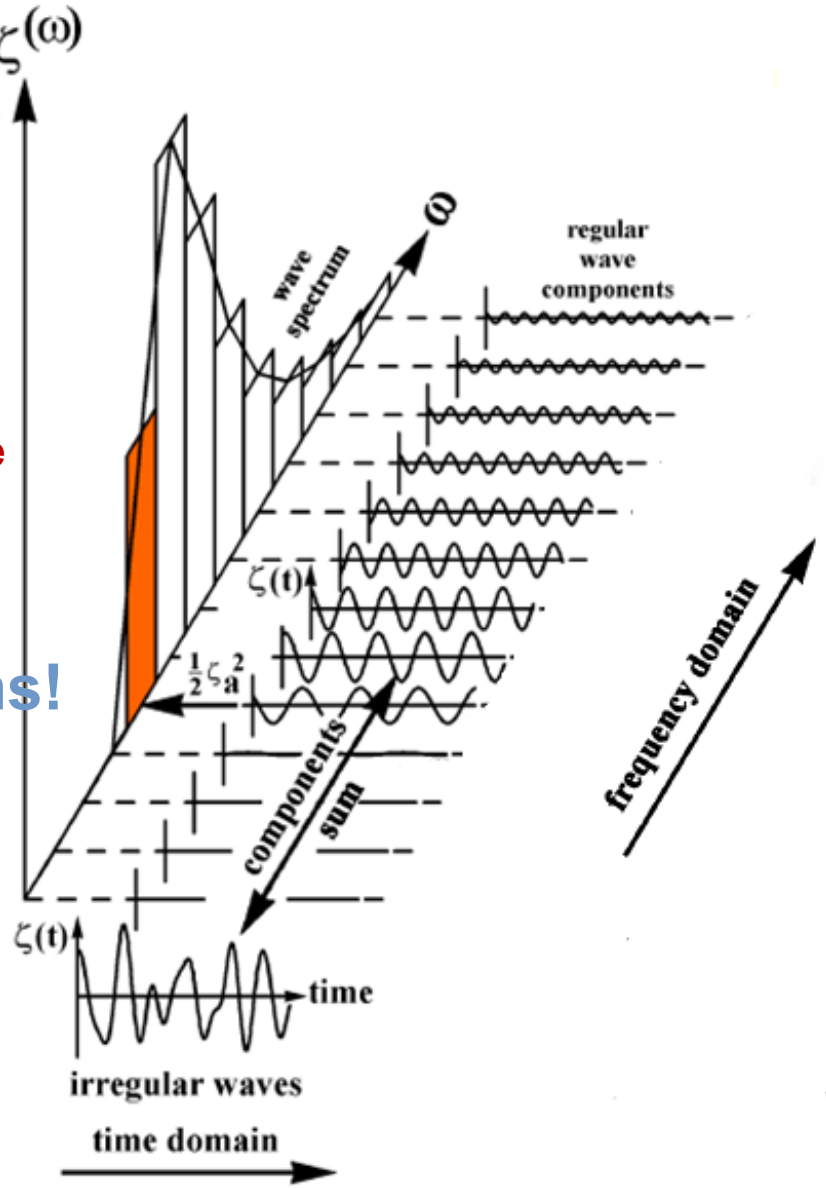
Sea state realisation

$$\eta(t) = \sum A_i \cdot \cos(\omega_i t - \varphi_i)$$

$$S(\omega)\, d\omega = \frac{A_i^2}{2}$$

Random phase from a uniform distribution

Infinite number of realisations!



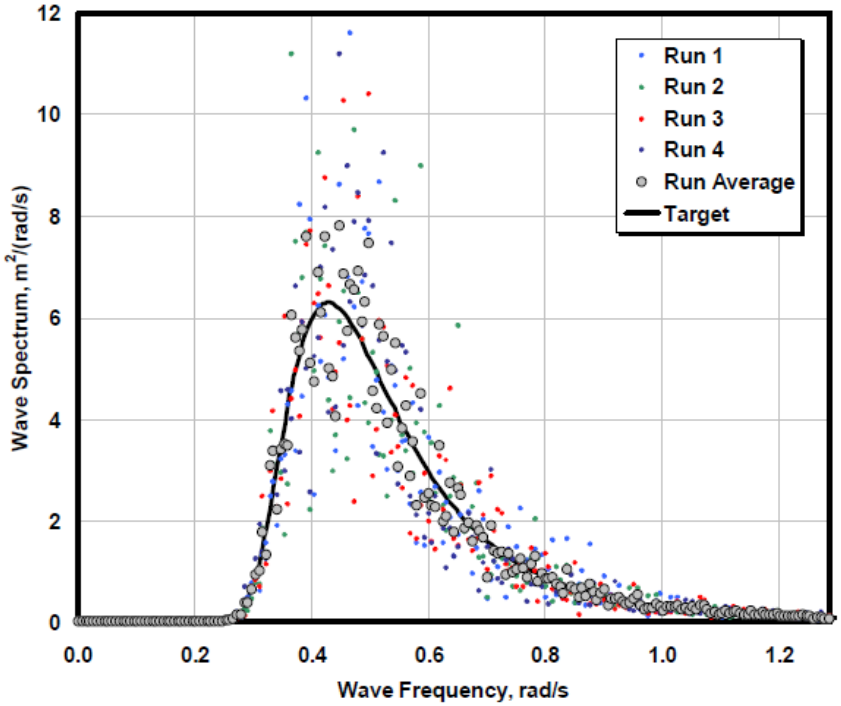
The wave phase only impacts the wave-by-wave analysis (Hmax, …)

Spectral analysis

Sea state realisation

Wave spectrum typically derived from in situ observations, or from a model calibrated/validated with observations

Spectrum is an average, which can be taken into account in a model by applying an amplitude distribution

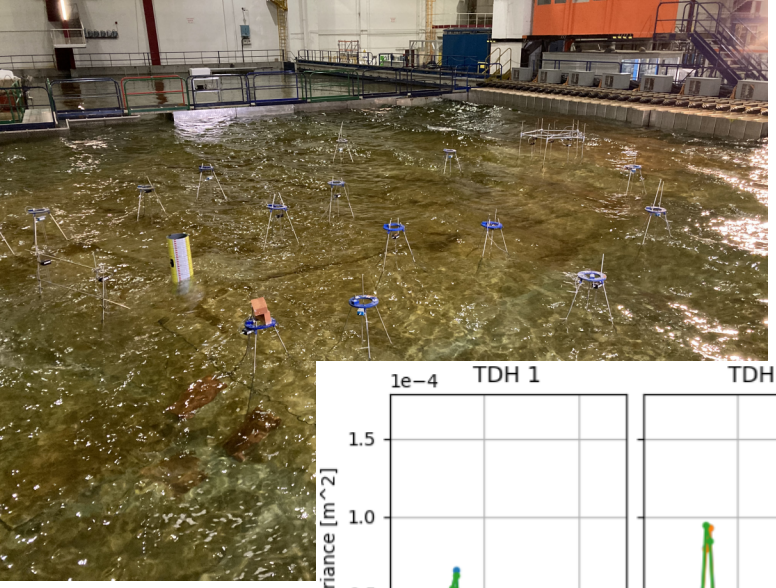


$$\eta(t) = \sum a_i \cdot \cos(\omega_i t + \varphi_i)$$

$$p(a_i) = \frac{\pi}{2}\frac{a_i}{{\mu_i}^2} \cdot \exp\left(-\pi {a_i}^2 \Big/ 4{\mu_i}^2\right)$$

$$\mu_i = \mathrm{E}(a_i) = \sqrt{2\mathrm{S}(\omega_i)\, d\omega_i}$$

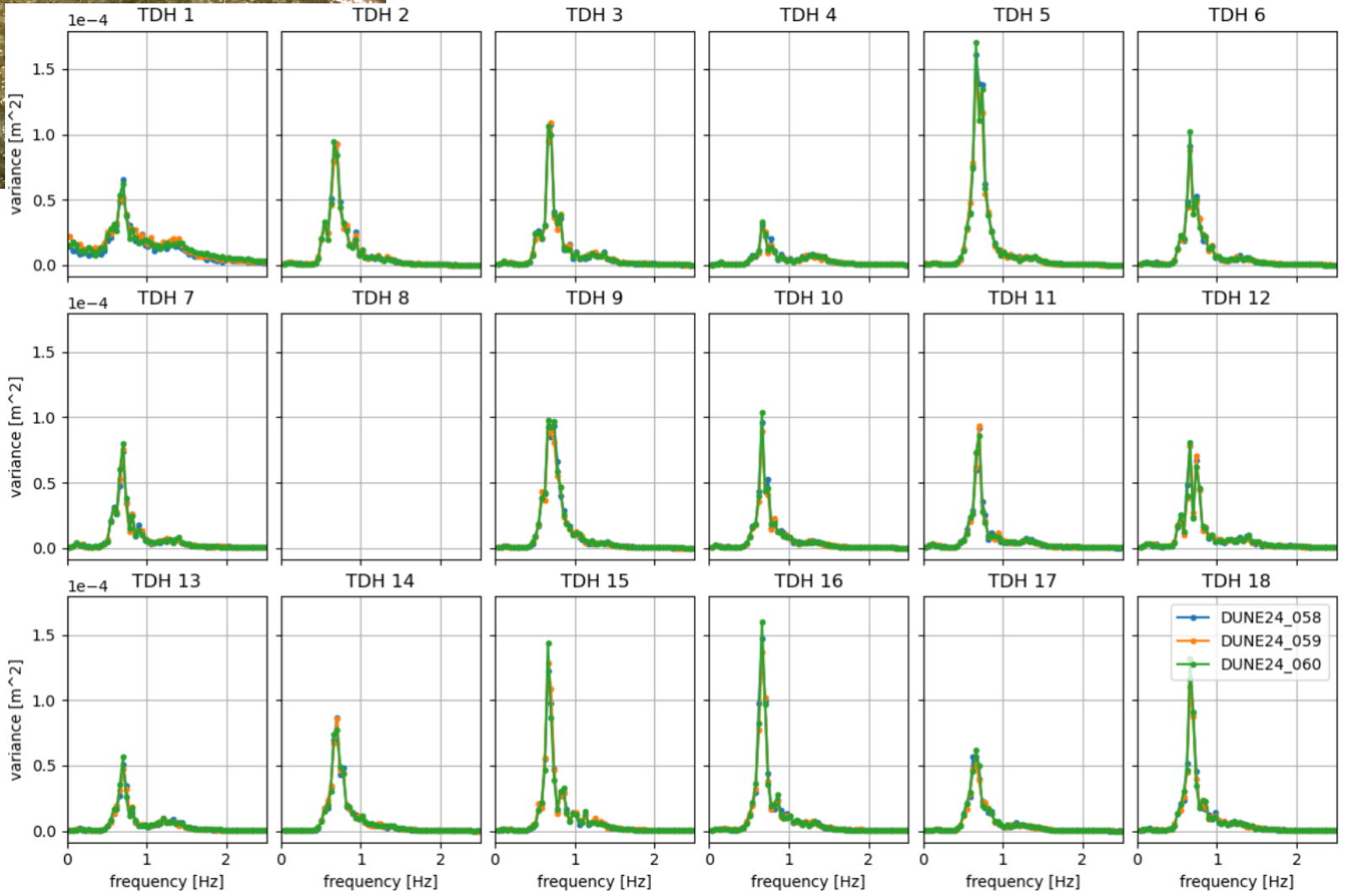
➡ Impacts on structure design!



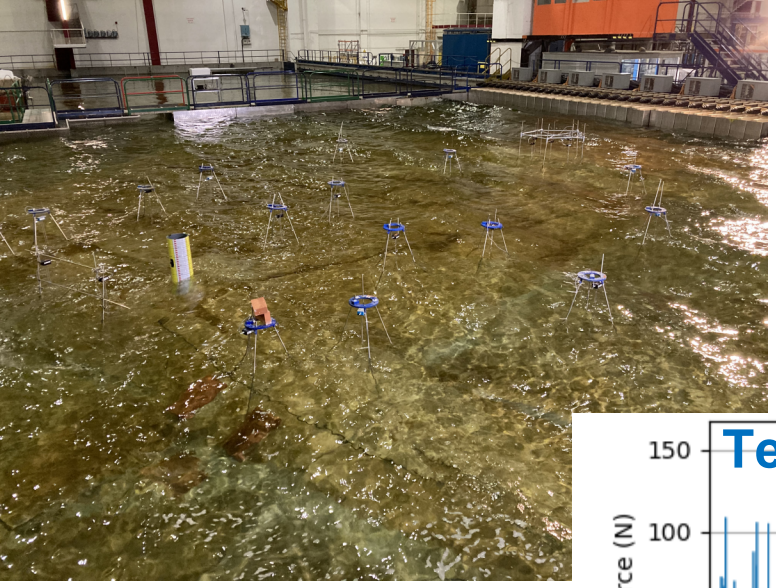
Laboratory Experiments

LABORATOIRE D'HYDRAULIQUE SAINT-VENANT

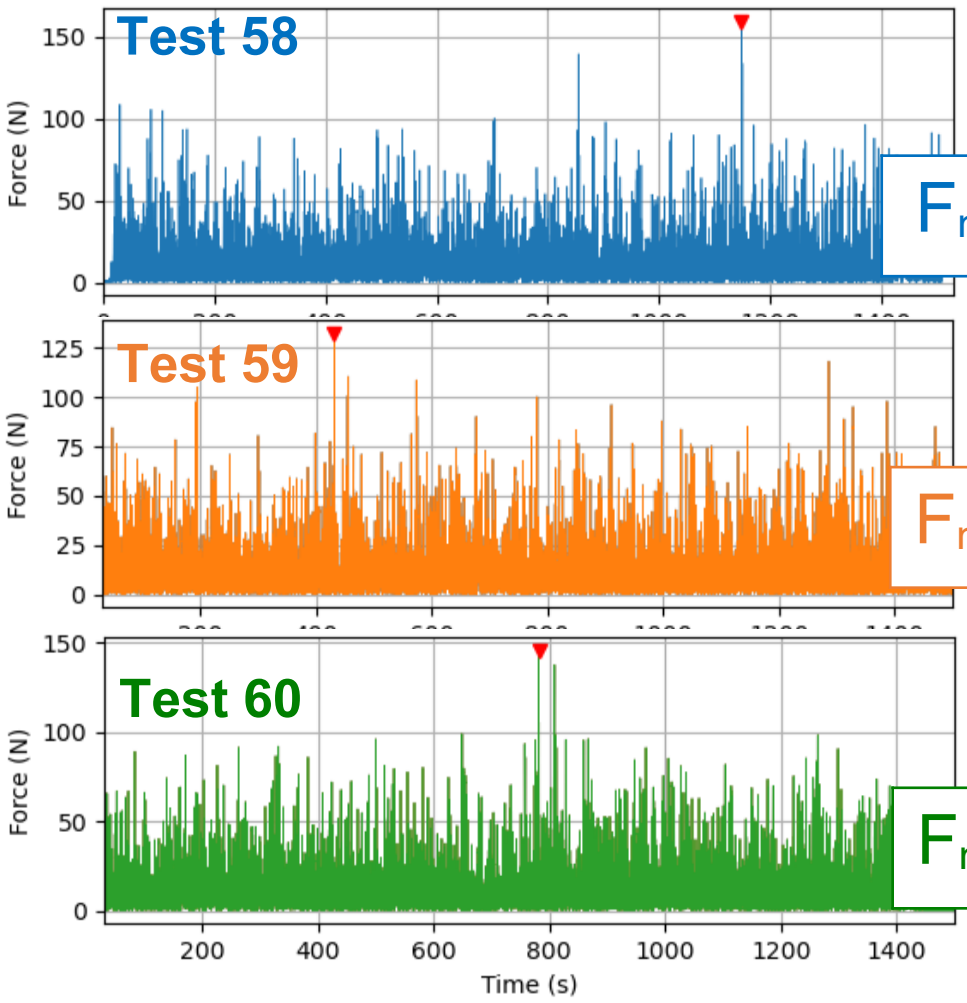
3 runs with the same wave spectrum but different seed



Laboratory Experiments



3 runs with the same wave spectrum but different seed



F_{max}=159 N

F_{max}= 132 N

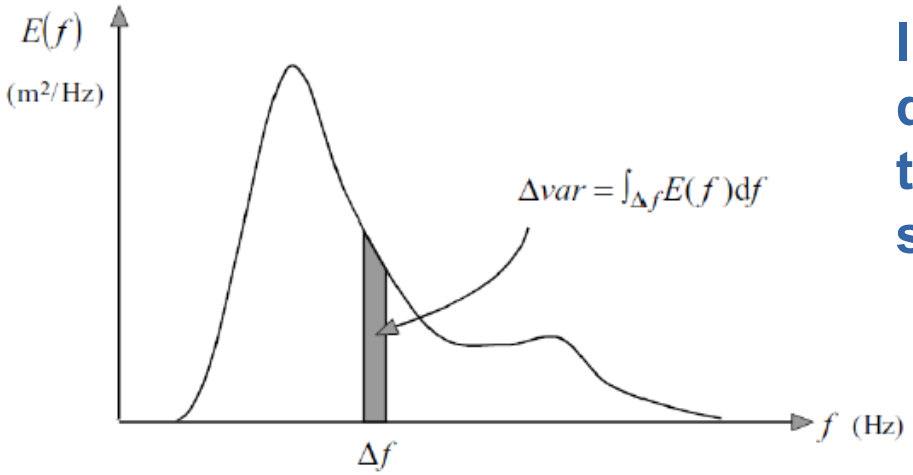
F_{max}= 145 N

Spectral analysis

Important quantities calculated from the wave spectrum:

Zeroth order moment

$$m_0 = \int_0^{+\infty} S(\omega)\ d\omega = \mathrm{R}(0)$$



Integration of the power density spectrum = total energy = free surface variance E(η²)

Nth order moment:

$$m_n = \int_0^{+\infty} S(\omega) \cdot \omega^n\ d\omega$$

(Holthuijsen, 2007)

Spectral analysis

Important quantities calculated from the wave spectrum:

Significant wave height H_{m0} , or H_s

$$Hm_0 = 4\sqrt{m_0} \approx H_{1/3}$$

= mean of 1/3 largest waves

Mean energy wave height H_E

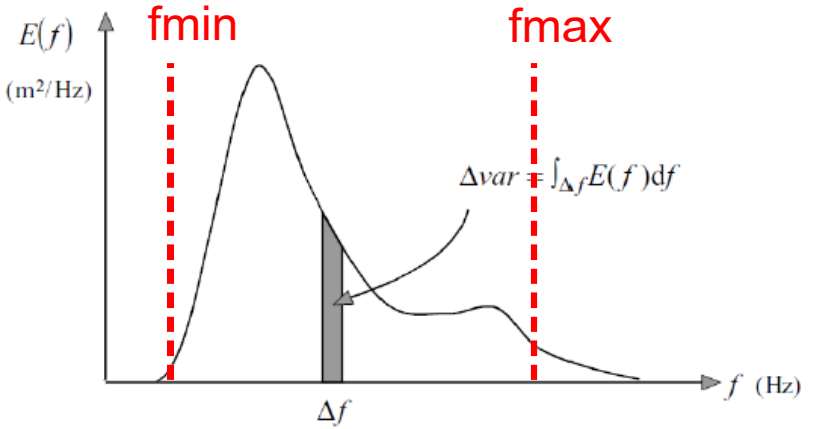
$$H_E = \sqrt{8m_0}$$

$$= \frac{1}{\sqrt{2}} H_s$$

= wave height of regular wave with the same energy as the irregular wave spectrum

Important:

- discretization (~100 freq.)
- cutoffs:
  - fmin = 0.4-0.5 fp
  - fmax = 1/(2Δt) = Nyquist freq.



(Holthuijsen, 2007)

Spectral analysis

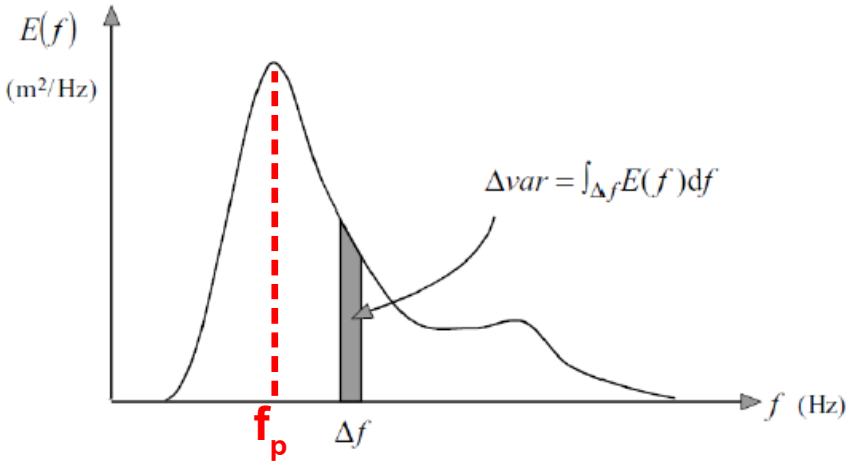
Important quantities calculated from the wave spectrum:

Peak period T_p , with $f_p=1/T_p$

= period at amplitude maximum

= estimated as:

$$f_p = \frac{\int_0^{+\infty} \{S(f)\}^n \cdot f \; df}{\int_0^{+\infty} \{S(f)\}^n \, df}$$



Mean wave period T_{01} , with $f_{01}=1/T_{01}$

= mean weighted by the frequency

$$T_{01} = \frac{m_0}{m_1}$$

Mean wave period T_{02} , avec $f_{02}=1/T_{02}$

= mean weighted by the frequency squared

$$T_{02} = \sqrt{\frac{m_0}{m_2}}$$

Mean wave period T_{-10} , avec $f_{-10}=1/T_{-10}$

= mean weighted by the period

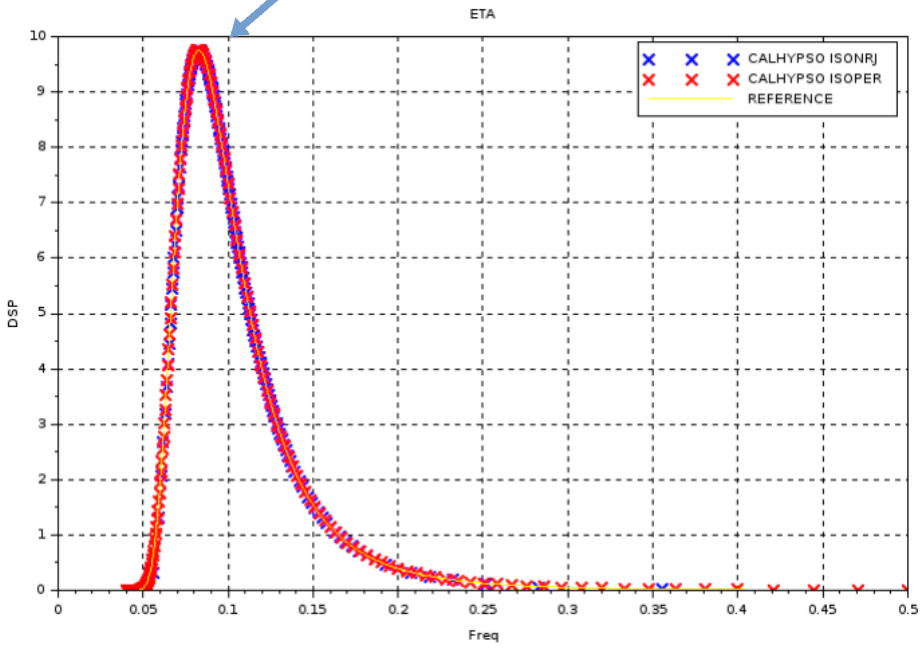
$$T_E = T_{-10} = \frac{m_{-1}}{m_0}$$

Spectral analysis

Commonly used models of wave spectra

Pierson-Moskowitz spectrum (« fully developped » seas)

$$S(f) = \frac{1}{4\pi} {H_s}^2 \left(\frac{2\pi}{T_Z}\right)^4 \omega^{-5} exp\left\{-\frac{1}{\pi}\left(\frac{2\pi}{T_Z}\right)^4 \omega^{-4}\right\}$$



- 2 parameters: Hs, Tz

$$T_p = 1{,}408\, T_Z$$

(up-crossing mean period = T_{02})

- developed from observations in the North Atlantic

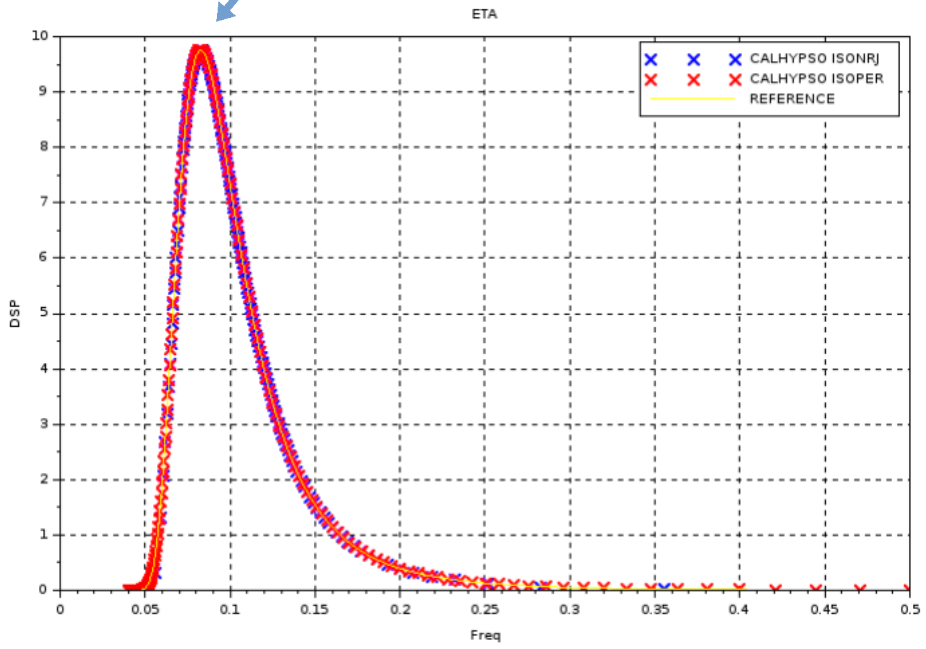
Spectral analysis

Commonly used models of wave spectra

JONSWAP spectrum (« developing » or « fetch-limited » seas)

$$S(f) = \delta H_s{}^2 f_p{}^4 f^{-5} \exp\left\{-1{,}25\left(f_p/f\right)^4\right\} \gamma^{exp\left\{-0{,}5\frac{\left(f/f_p - 1\right)^2}{\sigma^2}\right\}}$$

$$\sigma = 0{,}07 \text{ if } f < f_p\,; 0{,}09$$

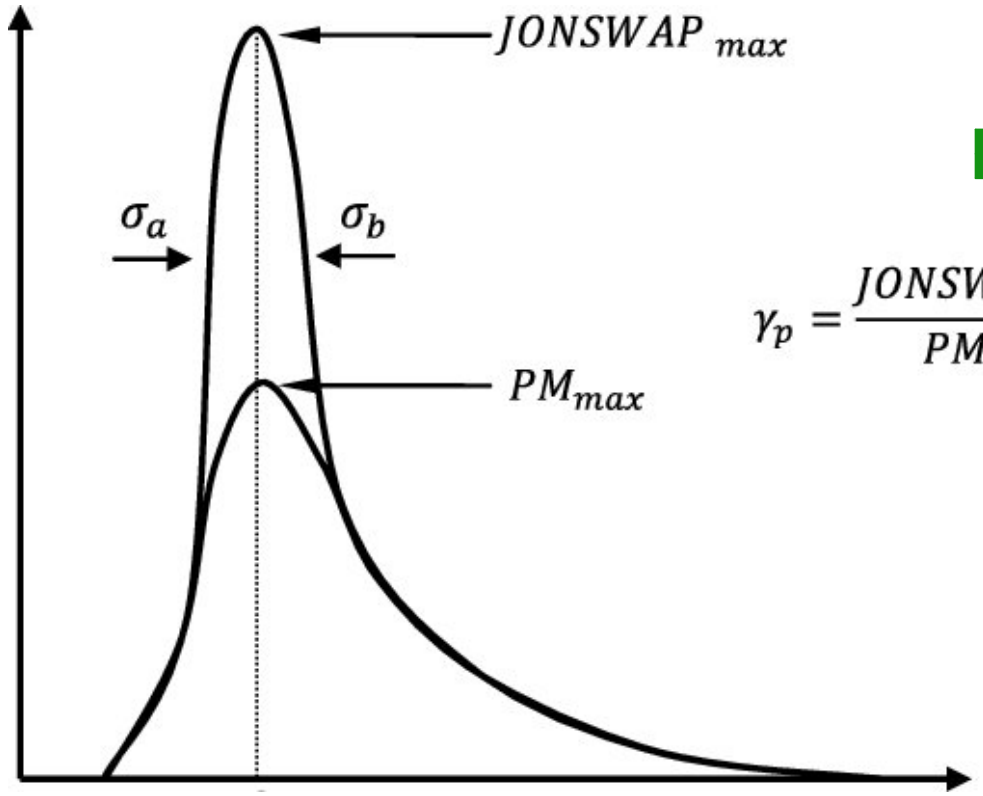


- 3 (4) parameters: H_s, f_p, γ, (σ)
- Decreases as f^{-5} in high frequency range
- JOint North Sea WAve Project: developed from observations in the North Sea

Spectral analysis

Commonly used models of wave spectra

Pierson-Moskowitz vs. JONSWAP spectrum



(Lambert, 2013)

$$\gamma_p = \frac{JONSWAP_{max}}{PM_{max}}$$

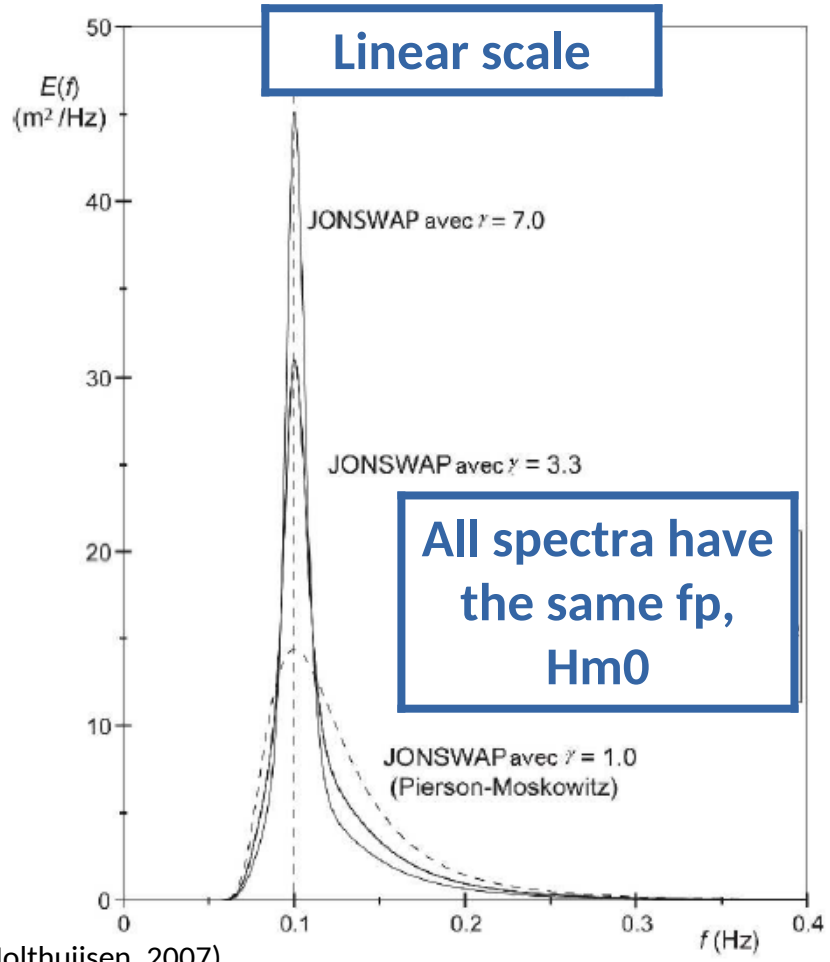
$$\gamma = 1$$

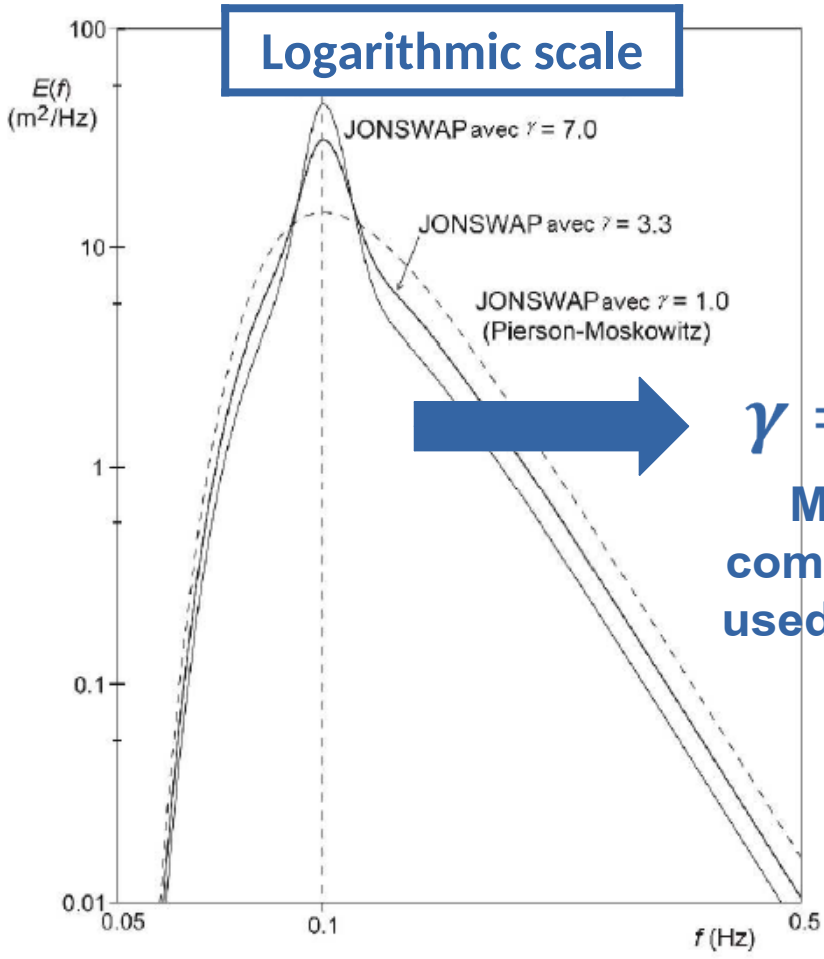
JONSWAP spectrum simplifies to P-M spectrum

Spectral analysis

Commonly used models of wave spectra

JONSWAP spectrum: effect of gamma factor (γ)





$\gamma = 3.3$

Most commonly used value

(Holthuijsen, 2007)

Spectral analysis

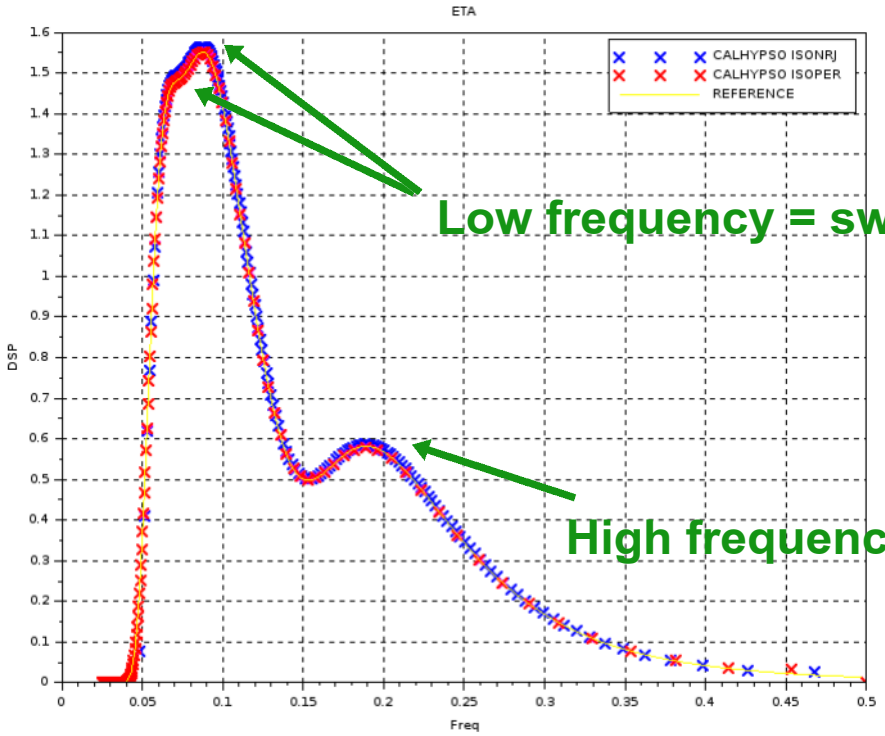
Commonly used models of wave spectra

Hochi-Hubble spectrum

$$S_v(f) = \frac{\left[(4\lambda_V + 1)\cdot f_V^{\,4}\right]^{\lambda_V} H_V^{\,2}}{4^{\lambda_V+1}\,\Gamma(\lambda_V)\cdot f^{\,4\lambda_V+1}} \exp\left[-\frac{4\lambda_V+1}{4}\left(\frac{f_V}{f}\right)^4\right]$$

$$S(f) = \sum_{v=1}^{3} S_v(f)$$

Superposition of 3 JONSWAP spectra



Low frequency = swell

High frequency = wind waves

- 9 parameters

Spectral analysis

Relationship between commonly used parameters

JONSWAP spectrum (most frequently used)

Up-crossing mean period versus peak period:

$$\frac{T_Z}{T_p} = 0{,}6063 + 0{,}1164 \cdot \gamma^{1/2} - 0{,}01224 \cdot \gamma$$

Mean period $T_{m\text{-}1,0}$ versus peak period

	Indice d'accroissement du pic γ du spectre de JONSWAP								
	1 (PM)	**2**	**3**	**3.3**	**5**	**7**	**10**	**15**	**20**
$T_p/T_{m\text{-}1,0}$	1.167	1.132	1.112	1.107	1.088	1.074	1.061	1.048	1.040

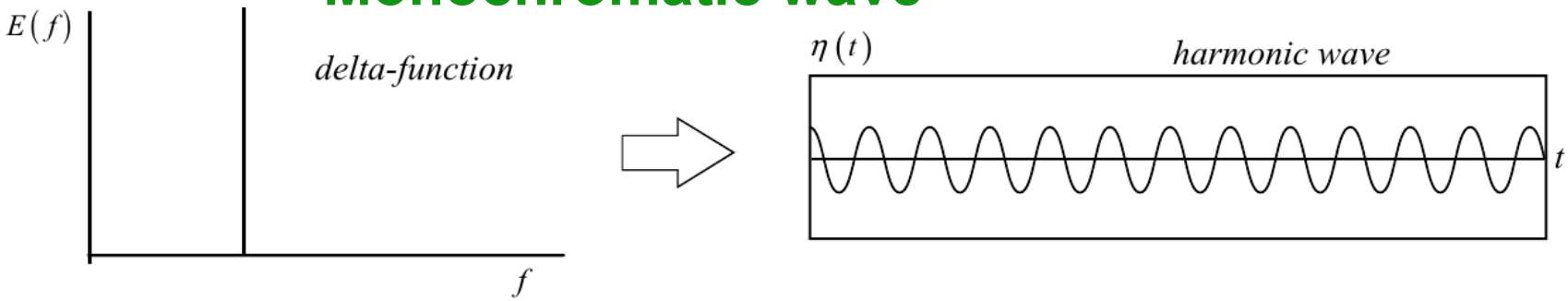
For a 3h sea state, *on average*:

$$\frac{H_{max}}{H_s} = 1{,}86$$ **(typically ranging from 1.6 to 2 in coastal zones)**

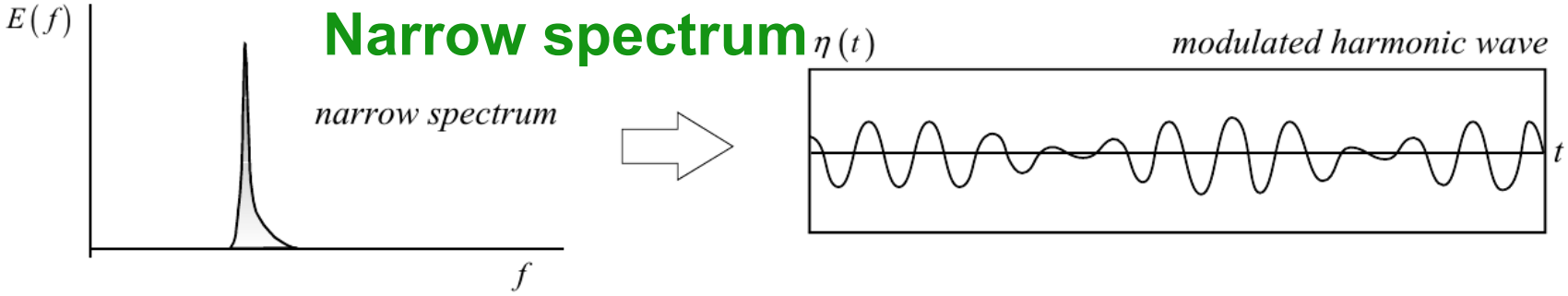
Spectral analysis

Spectral shapes

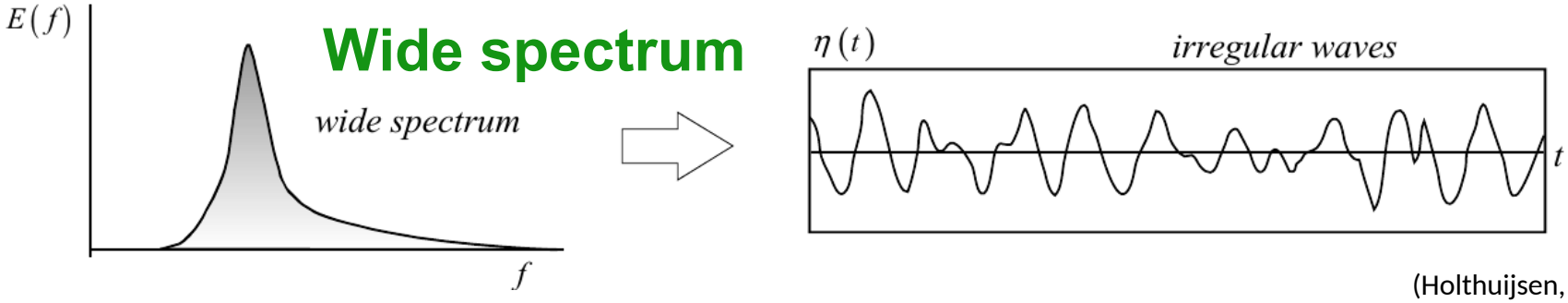
Monochromatic wave



Narrow spectrum



Wide spectrum

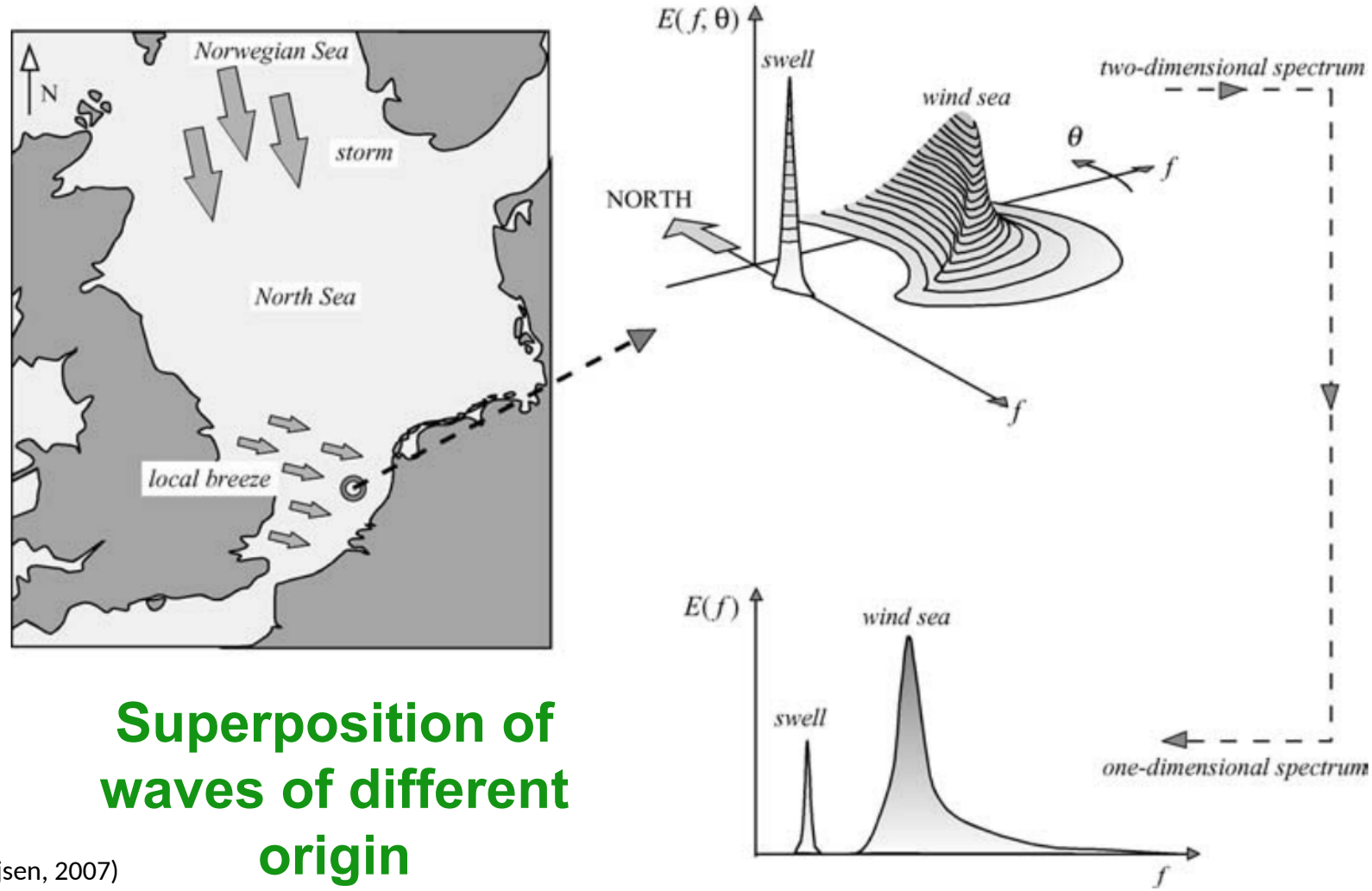


(Holthuijsen, 2007)

Spectral analysis

Spectral shapes

Multi-peak spectra



Superposition of waves of different origin

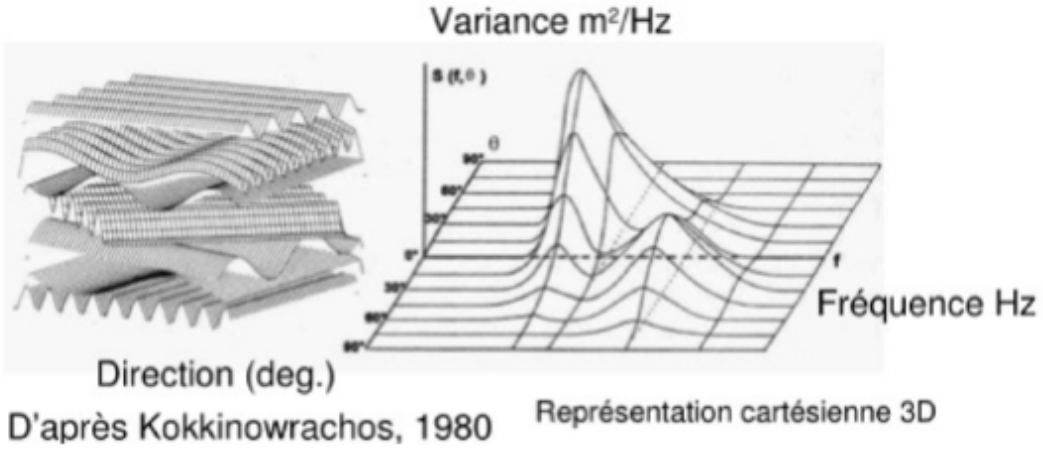
(Holthuijsen, 2007)

Spectral analysis

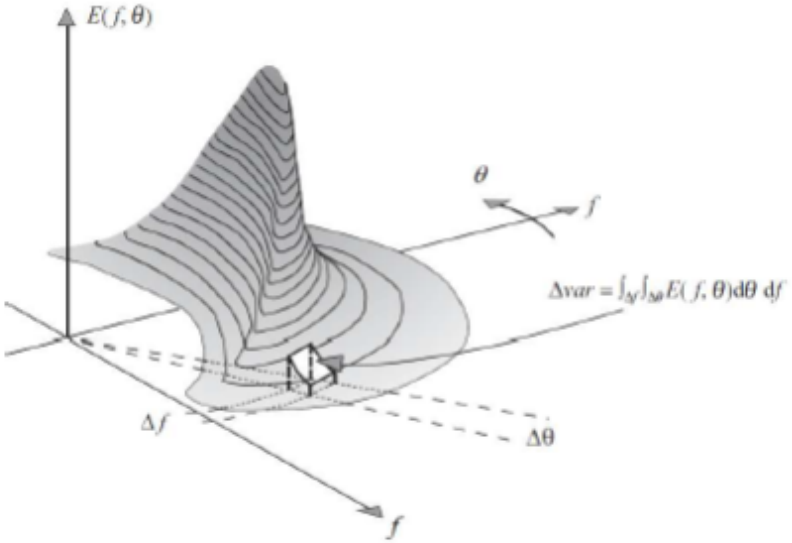
Unidirectional ⟶ Multidirectional spectrum:

Superimposing monochromatic waves (Gaussian model) with different incident directions

$$\eta(t) = \sum A_i \cdot \cos(k_i cos(\theta_i)x + k_i sin(\theta_i)y - \omega_i t - \varphi_i)$$



2D spectrum (f , θ)



(Holthuijsen, 2007)

Spectral analysis

Multidirectional spectrum:

Directional spreading given analytically :

$$D(f,\theta) \qquad \int_0^{2\pi} D(f,\theta)d\theta=1 \quad \Rightarrow \quad \hat{S}(f,\theta) = S(f)\cdot D(f,\theta)$$

$$\Rightarrow \quad \iint \hat{S}(f,\theta)df\,d\theta = \int S(f)\,df$$

Example (Mitsuyasu 1975)

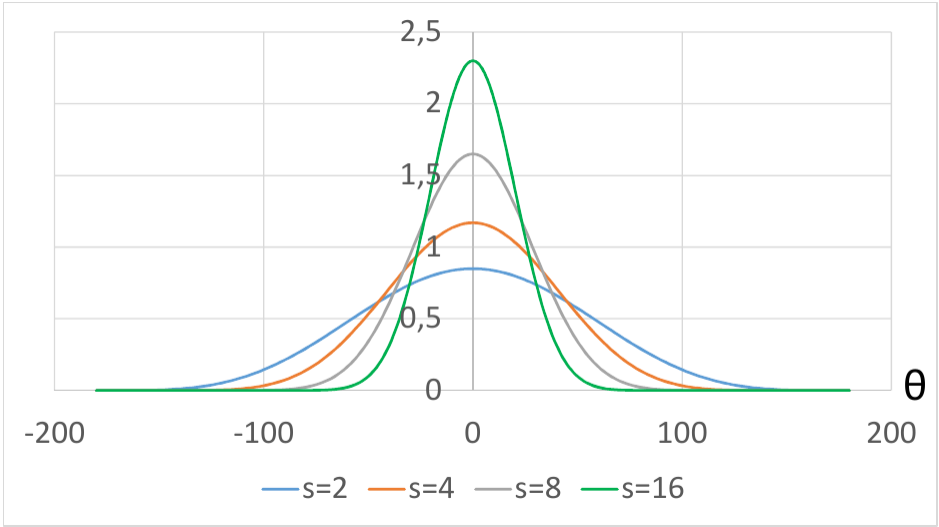
$$D(f,\theta) = C_1(s)\cdot cos^{2s}\left(\frac{\theta-\theta_0}{2}\right)$$

$$-\pi < \theta - \theta_0 < \pi$$

Others exist…

$$D(f,\theta) = C_2(s)\cdot cos^{2s}(\theta-\theta_0)$$

$$-\pi/2 < |\theta - \theta_0| < \pi/2$$

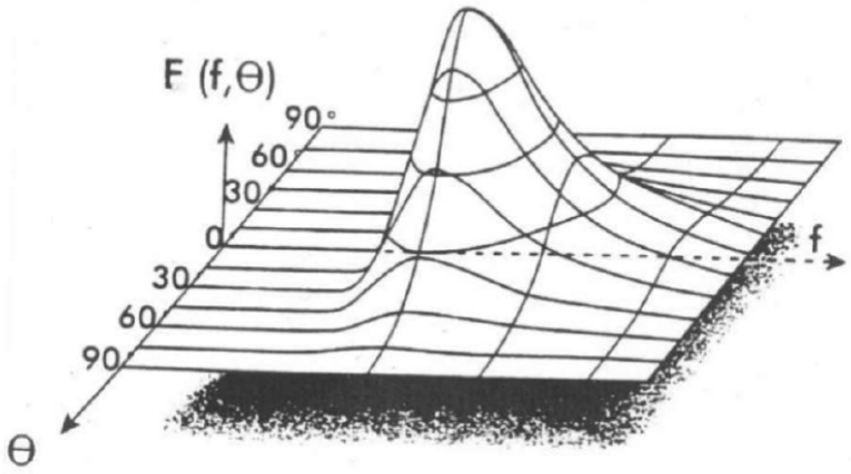


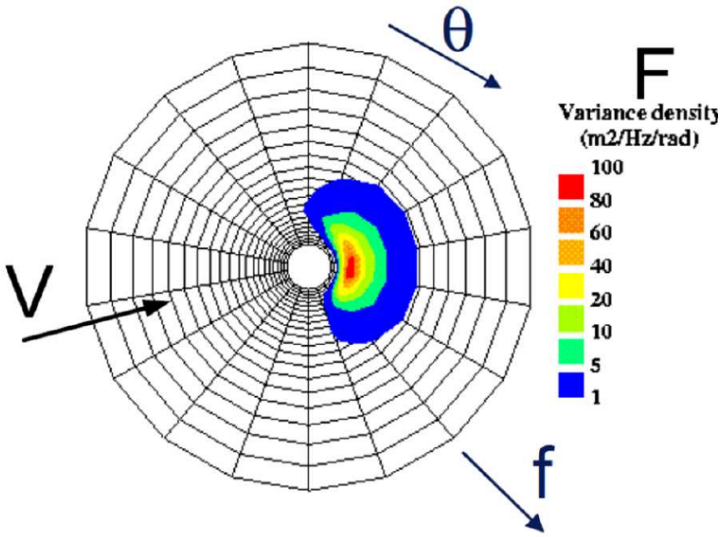
Spectral analysis

Directional spectrum

Cartesian (2 orthogonal

Polar (isolines)





Typically use variance spectrum F(f, θ) in data analysis and modeling:

$$F(f,\theta) = \frac{1}{\rho g} S(f,\theta) = E(f).D(f,\theta)$$

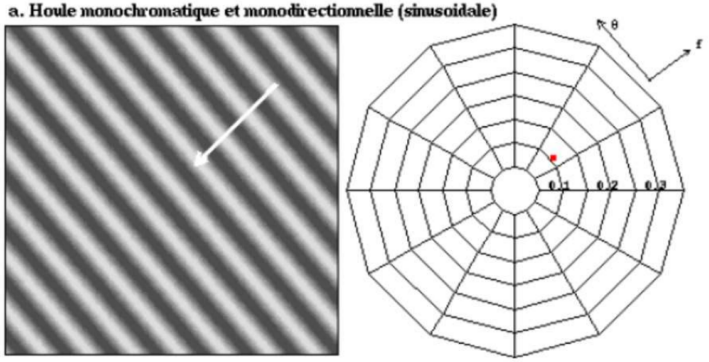
$$F(f) = \int_{\theta=0}^{2\pi} F(f,\theta)\,d\theta$$

[m2/Hz/rad]

Spectral analysis

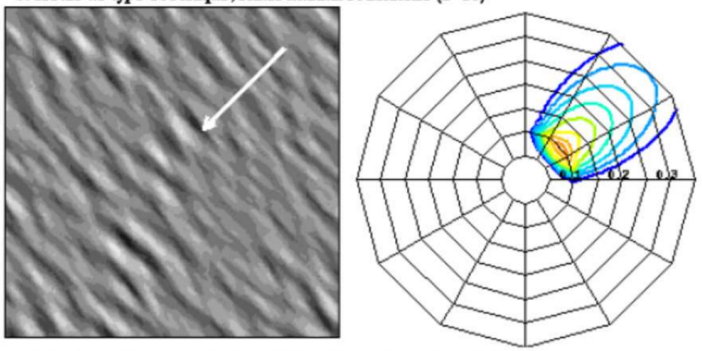
Directional spectrum

a. Houle monochromatique et monodirectionnelle (sinusoidale)



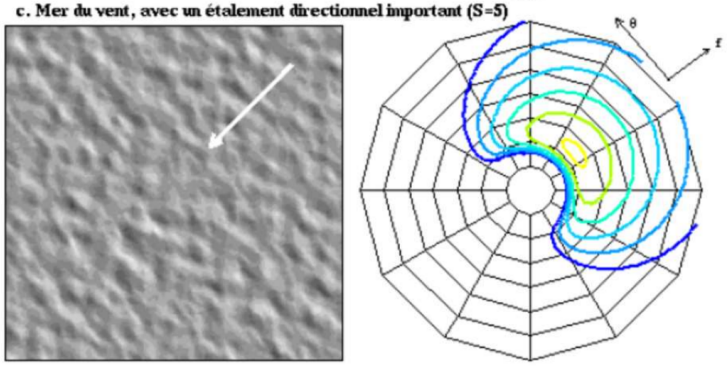
Monochromatic and unidirectional (1 component)

b. Houle de type océanique, faible multidirectionnelle (S=50)



Narrow spreading multidirectional waves (s=50)

c. Mer du vent, avec un étalement directionnel important (S=5)

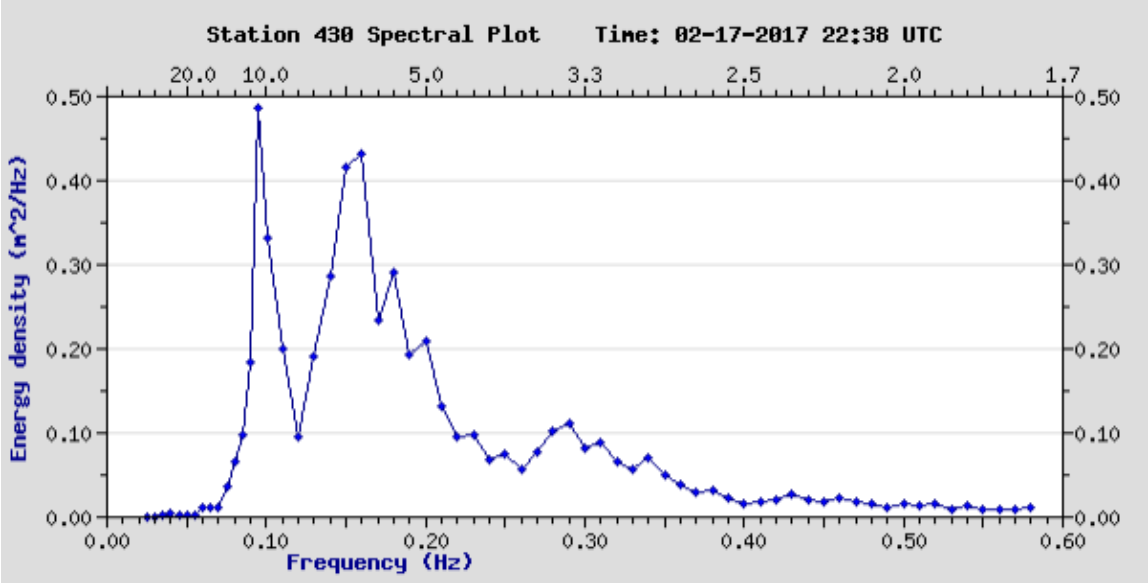


Wide spreading multidirectional waves (s=5)

Spectral analysis

Measured wave spectra

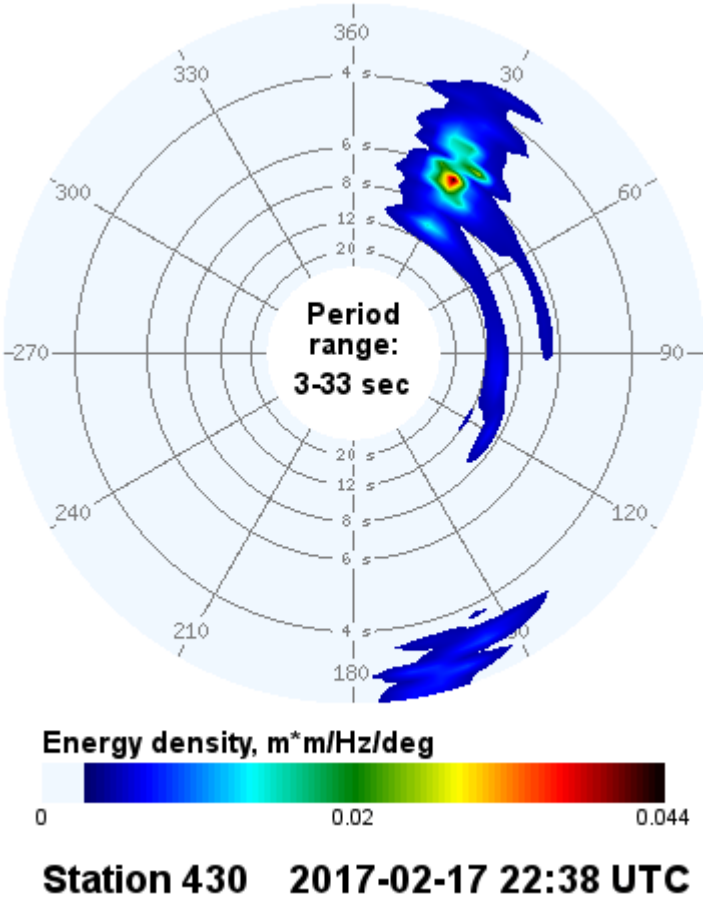
Frequency spectrum (1D)



Wave buoy: Duck, NC, USA (26m)
February 17, 2017

Source : cdip.ucsd.edu

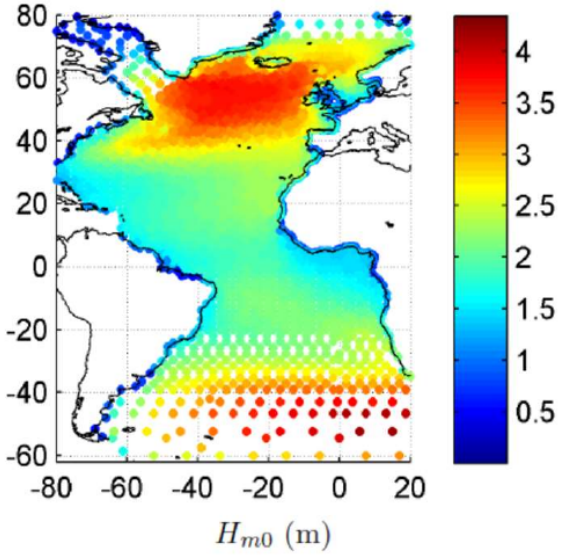
Rose diagram (2D: frequency, direction)

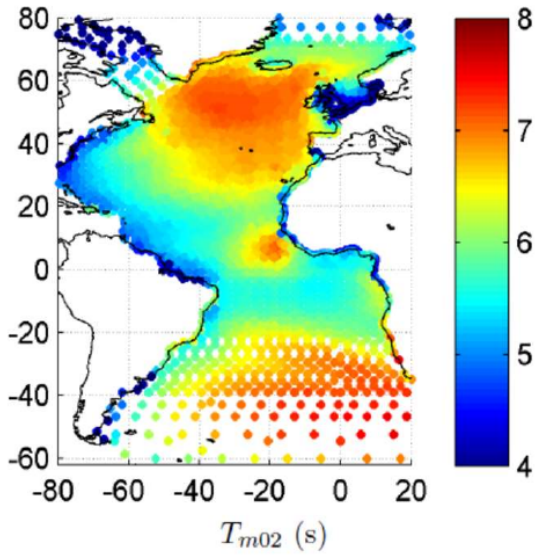


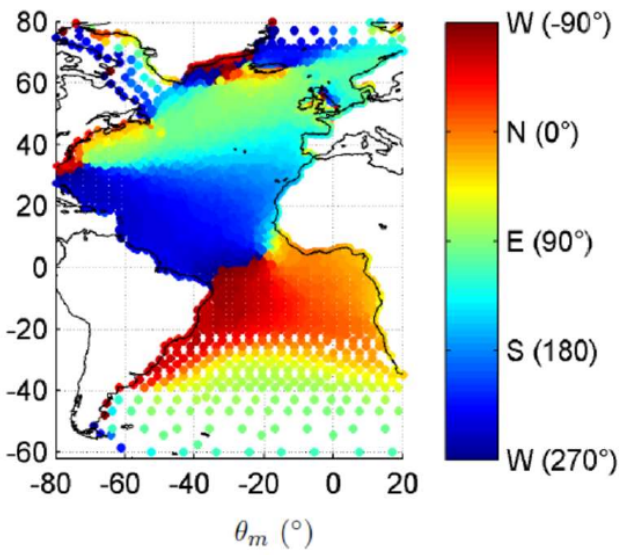
Wave climatologies

Historical

- In situ observations (long term)
- Numerical simulations using wind reanalysis products

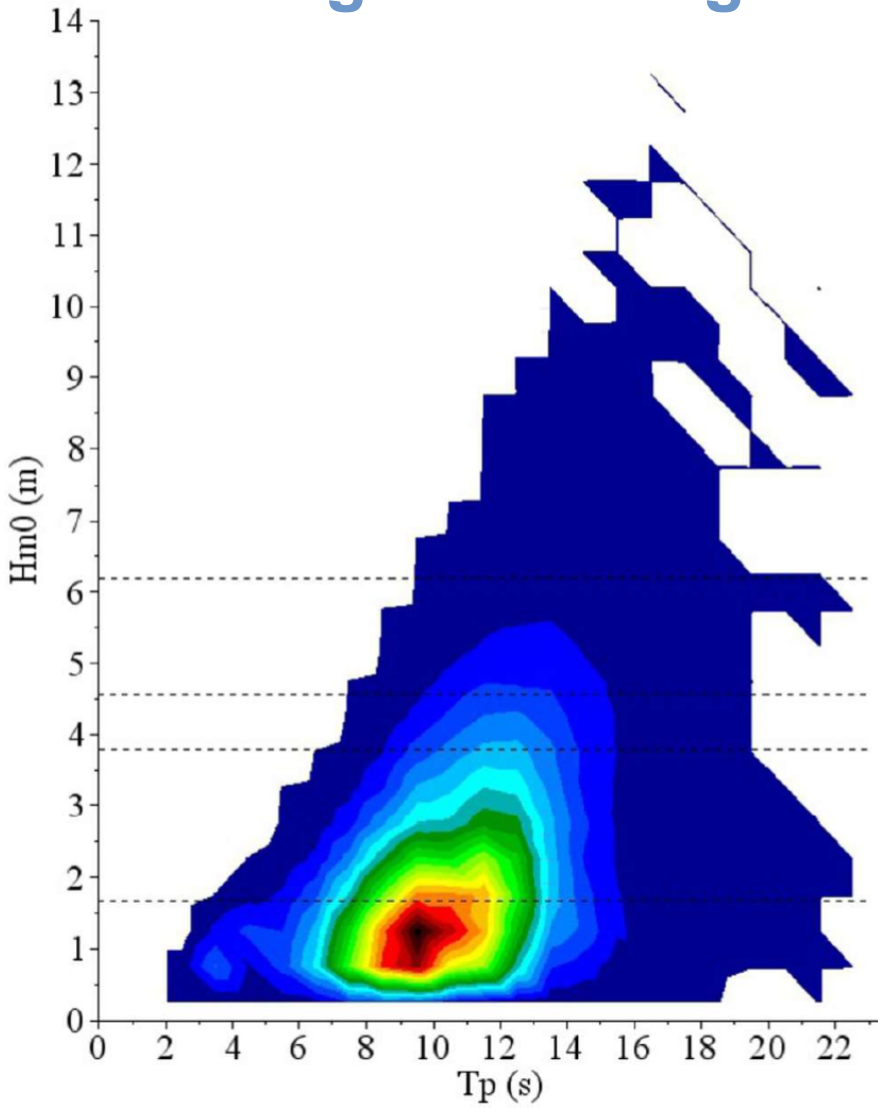






Wave climatology: In situ observations

Histogram of long-term statistics



Observation length:

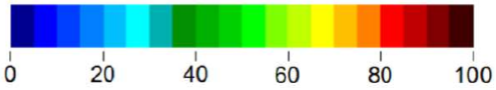
7.55 years in the period 15/10/2005 – 23/11/2015

e.g. in France:

Candhis

(https://candhis.cerema.fr/)

Normalized color scale



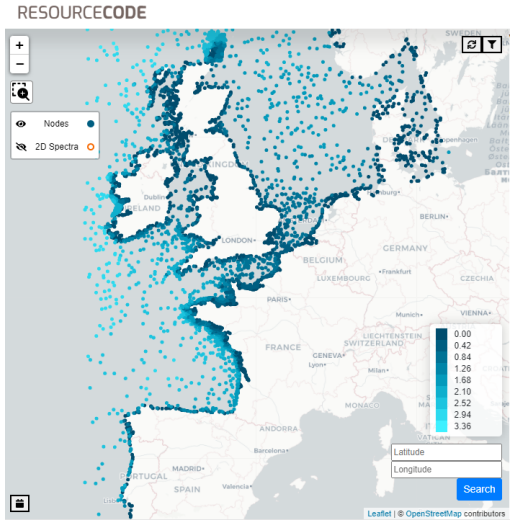
Wave climatology: numerical simulations

Hincasts: databases of wave statistics

e.g. in France:

RESOURCECODE

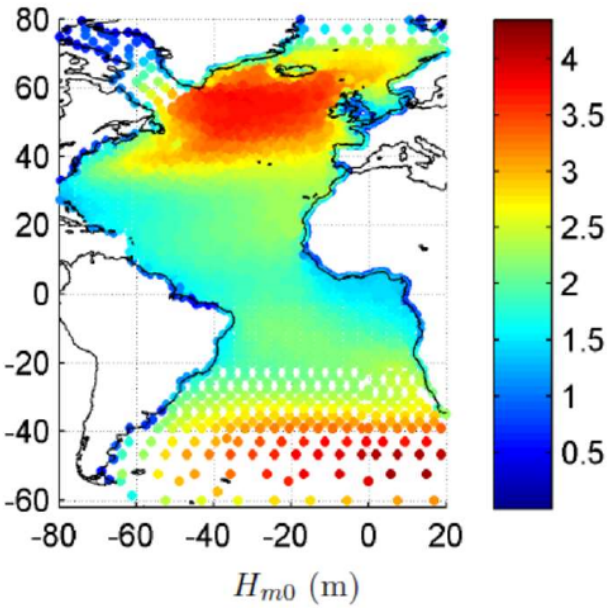
(https://resourcecode.ifremer.fr/)



ANEMOC

(https://www.cdata.cerema.fr/geonetwork/srv/fre/catalog.search;jsessi
id=A9AFE40E46F78DD96A3DD70543FB6835#/metadata/ef16f169-e208-
4549-948b-4b01c27d1508)

Example: Atlantic Ocean mean wave climate from 1979-2009, ANEMOC-2 (Laugel, 2013)



Wave climatology: numerical simulations

Hincasts: databases of wave statistics

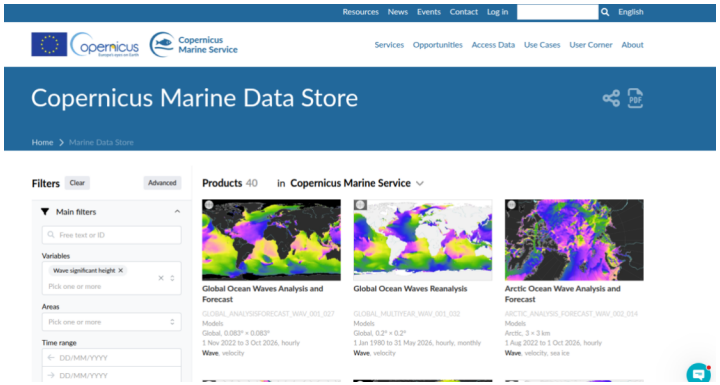
COWCLIP (Coordinated Ocean Wave Climate Project) : global ensembles

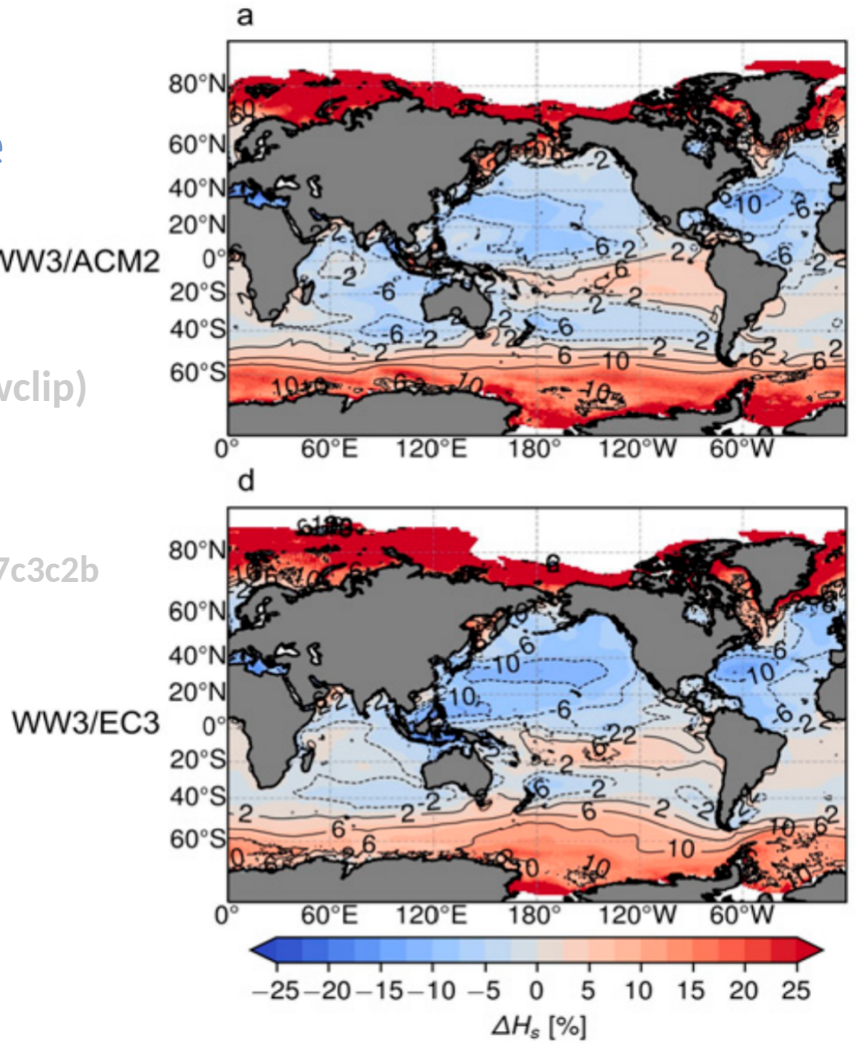
(https://community.wmo.int/site/knowledge-hub/programmes-and-initiatives/marine-services/coordinated-ocean-wave-climate-project-cowclip)

Copernicus Marine Data Store

(https://data.marine.copernicus.eu/products?pk_vid=681a5f7c3c2b8ee9179019716230c910)

Global and regional scale data sets





(Meucci et al., 2024, Fig. 17)

Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09
2. Linear wave theory	**Characterizing offshore waves**	**M. Yates**	**09/10**
3. Nearshore wave propagation	Calculating extreme wave conditions	M. Yates	16/10
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	23/10
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	06/11
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	04/12
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12
10. Final project presentations 13h30 - 17h15			***18/12***